



ĐẠI HỌC BÁCH KHOA HÀ NỘI
VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

Operating Systems

(Principles of Operating Systems)

Đỗ Quốc Huy

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Department of Computer Science

School of Information and Communication Technology

Course Info

Prerequisite:

- Introduction to information technology, Data structure and algorithm
- C programming skill

Course Info

Objectives:

- Know clearly the components of a computer system, problems and solutions in the history of computing system development
- Master the algorithms used in the OS, be able to evaluate algorithms, as well as be able to apply them to solve real-life problems
- Understand basic OS 'services (related to processes, threads, network, memory, directories, files)
- Grasp knowledge of data structures used in OS, improve concurrency programming capabilities, and system-level programming

Course Outline

Chapters:

- 1.Introduction (2.5 weeks)
- 2.Process Management (5.5 weeks)
- 3.Memory Management (3 weeks)
- 4.Files Systems (2 weeks)
- 5.IO Management (1 week)

Group Projects

Course Info

Text book:

- Operating System Concepts – Abraham Siblingschatz
- Modern Operating System – Andrew Tanenbaum
- Bài giảng Hệ Điều Hành – Nguyễn Thanh Tùng
- Nguyên lý Hệ Điều Hành – Đỗ Văn Uy
- Others universities' textbook

Class Expectations

Lectures

- Come! Cannot guarantee content will be identical to previous years.
- Meet your fellow students, they are your future teammate or colleagues!
- Electronic devices used only for **note taking**.

Exams

- Attendance mandatory

Communication Policy

- Communicate with lecturer, classmate through **Teams** rather than email

Chapter 1 Operating System Overview

- ① Operating system concept
- ② History of operating Systems
- ③ Definition and Classifications
- ④ Concepts in operating systems
- ⑤ Operating System structures
- ⑥ Basic Properties of Operating Systems
- ⑦ Principles of Operating Systems

Chapter 1 Operating System Overview

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Chapter 1. Operating System Overview

1. Operating system concept

① Operating system concept

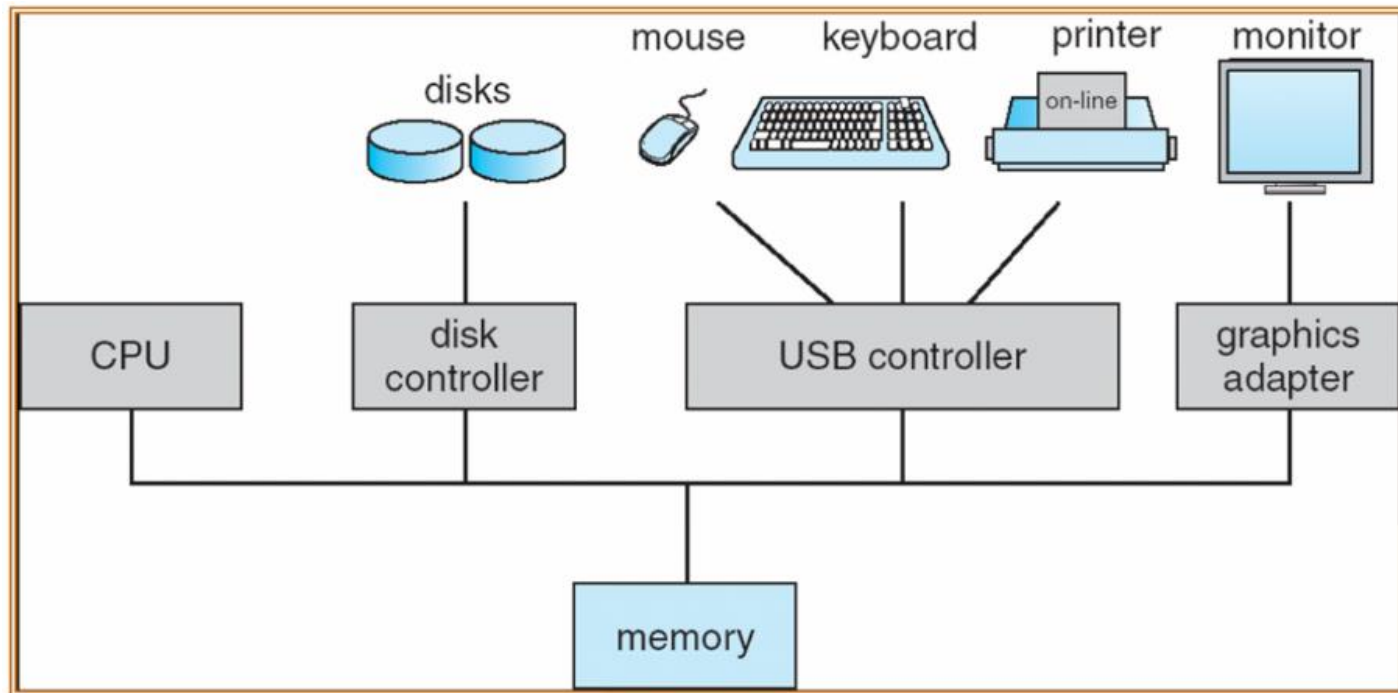
- Layered structure of a computing system
- Operating system's functions

Chapter 1. Operating System Overview

1. Operating system concept

1.1. Layered structure of a computing system

A computer system's structure (Silberschatz 2002)

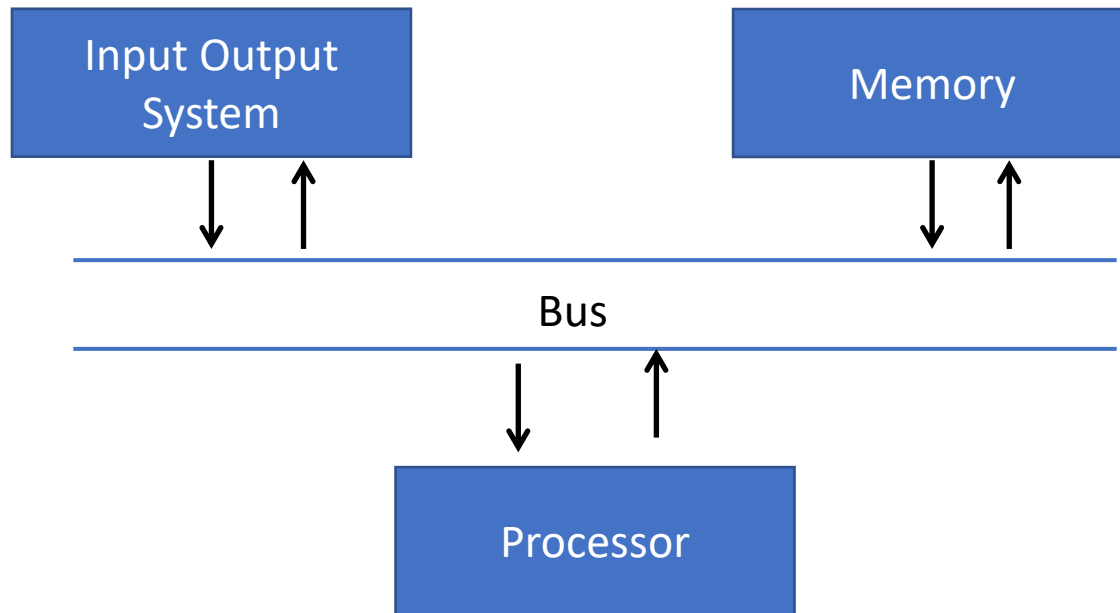


Chapter 1. Operating System Overview

1. Operating system concept

1.1. Layered structure of a computing system

A computer system's structure



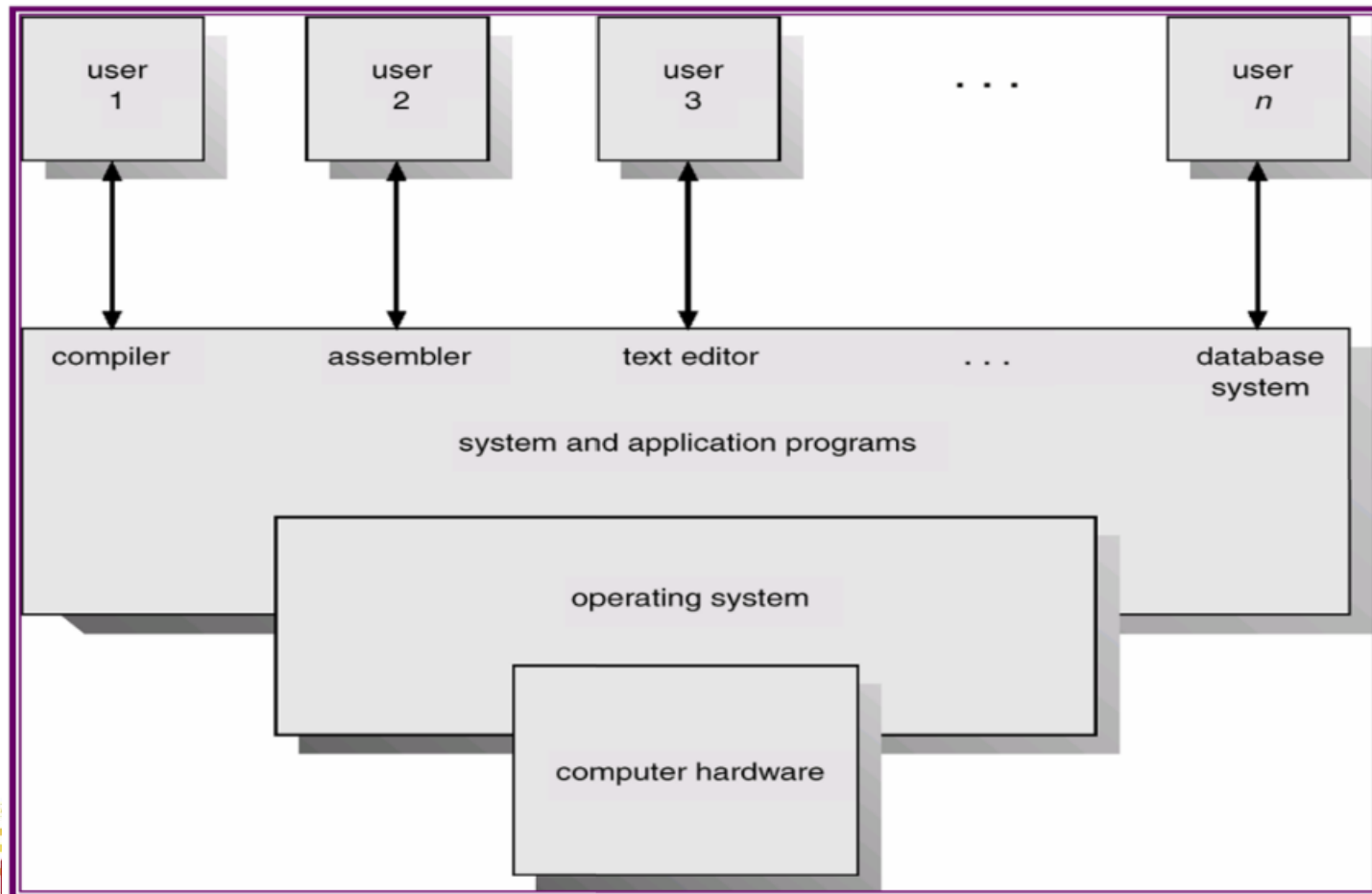
- One/ many CPUs, controlling devices are linked by a common bus system to access a shared memory.
- Controlling devices and CPU operate simultaneously and compete with each other.

Chapter 1. Operating System Overview

1. Operating system concept

1.1. Layered structure of a computing system

A computer system's components (Silberschatz 2002)

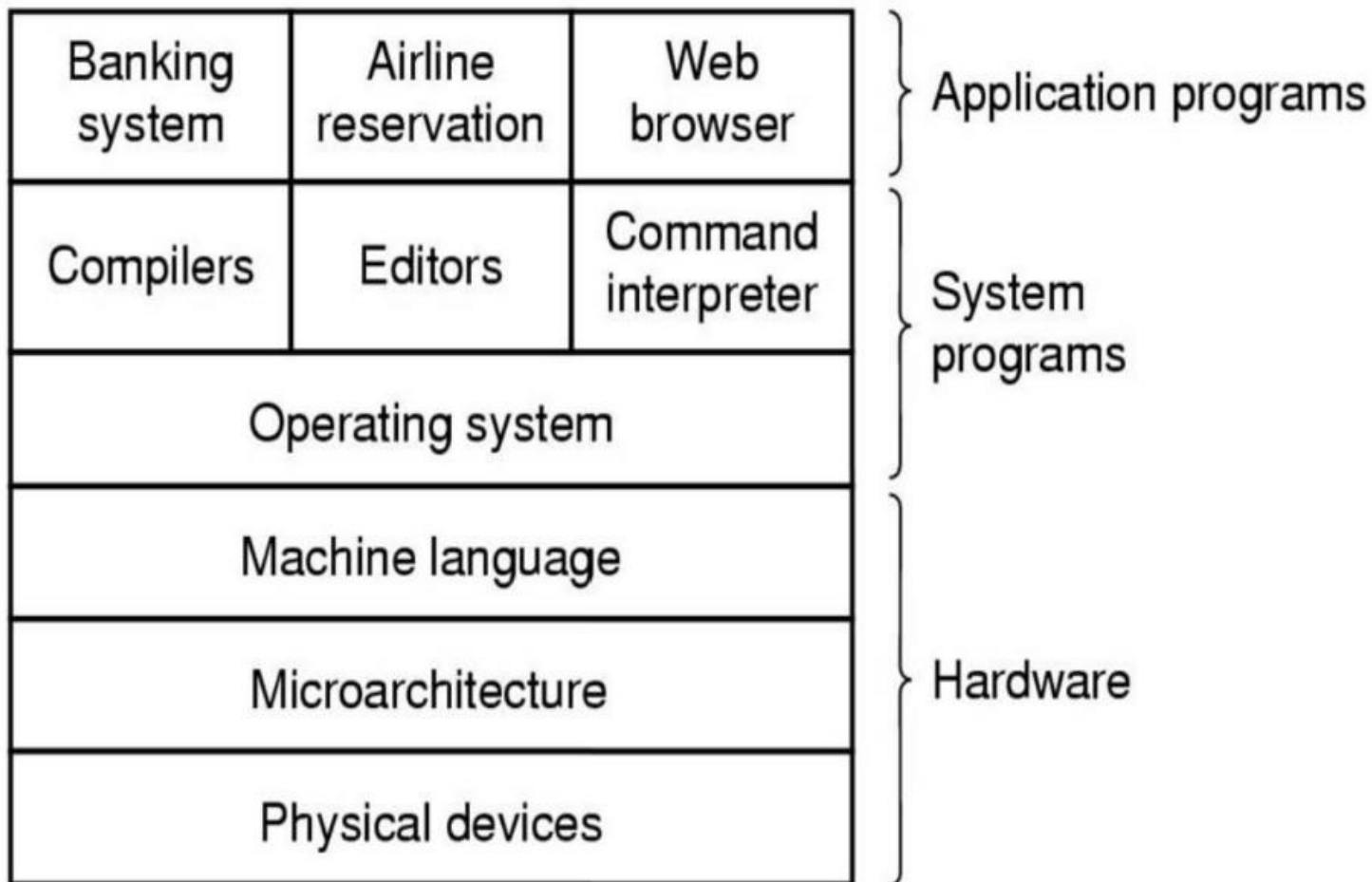


Chapter 1. Operating System Overview

1. Operating system concept

1.1. Layered structure of a computing system

A computer system's components (Tanenbaum 2001)



Chapter 1. Operating System Overview

1. Operating system concept

1.1. Layered structure of a computing system

A computing system's components

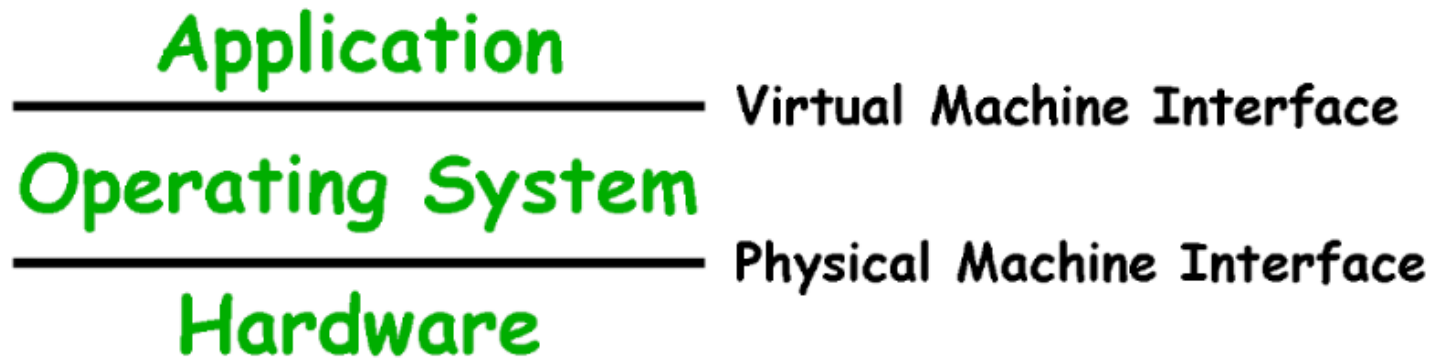
1. Hardware
2. Operating system
3. Applications programs
4. Users

Chapter 1. Operating System Overview

1. Operating system concept

1.1. Layered structure of a computing system

Objectives



- **Objectives:** To provide an **environment** which helps user **run application program** and **use** the computer system **easier**, more **conveniently** and **effectively**.
 - **Standardize** the user **interface** for **different hardware** systems
 - **Utilize** hardware's **resource effectively** and exploit the hardware's **performance optimally**.

Chapter 1. Operating System Overview

1. Operating system concept

① Operating system concept

- Layered structure of a computing system
- Operating system's functions

Chapter 1. Operating System Overview

1. Operating system concept

1.2. Operating system's functions

① Simulate a virtual computer machine

② Manage system resources

Chapter 1. Operating System Overview

1. Operating system concept

1.2. Operating system's functions

Simulate a virtual computer machine

Help **hide detailed works** and exploit computer hardware's functions easier and more effectively.

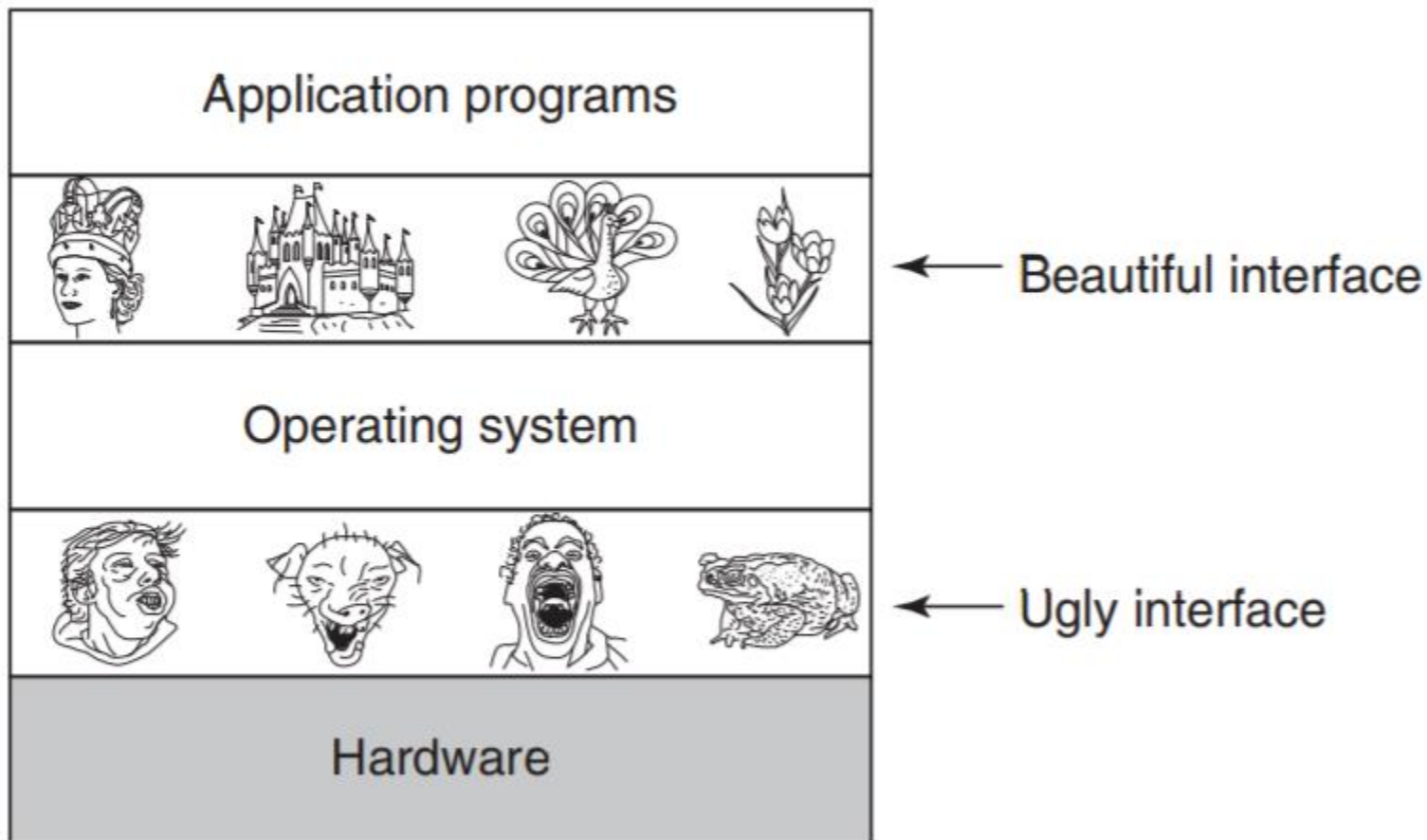
- Simplify programming problem
 - No need to work with binary sequences
 - Each program thinks that it own the whole computer's memory, CPU time, devices...
 - Help communicating with devices easier than with original device. Ex: Ethernet card: Reliable communication, ordered (TCP/IP)

Chapter 1. Operating System Overview

1. Operating system concept

1.2. Operating system's functions

Simulate a virtual computer machine



Chapter 1. Operating System Overview

1. Operating system concept

1.2. Operating system's functions

Simulate a virtual computer

- Extend the system's abilities: The system seem to have **desired resources** (virtual memory, virtual printer...)
- Help programs not violating each other → a program which does not work would not damage the whole system
- Useful for OS's development
 - If experimental OS get errors, only limited in the virtual machine
 - Help verifying other programs in the OS

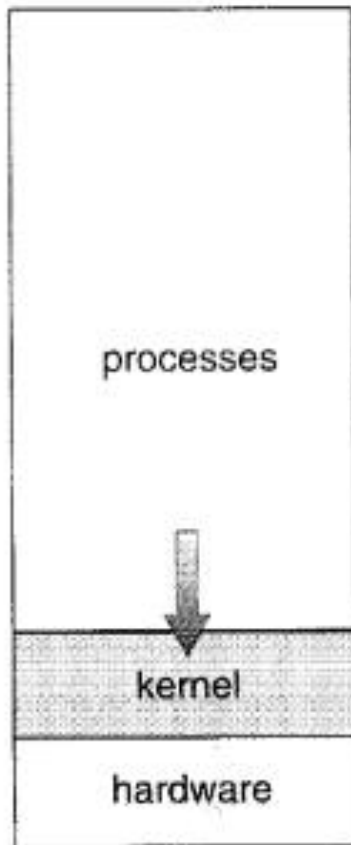
Chapter 1. Operating System Overview

1. Definition of operating system

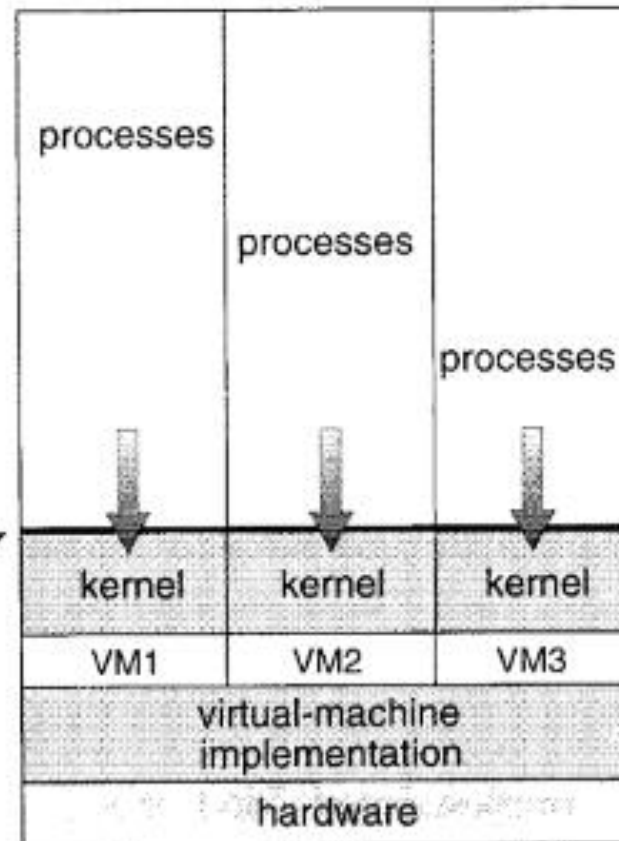
1.2. Operating system's functions

Simulate a virtual computer machine

Non virtual machine



With virtual machine



Chapter 1. Operating System Overview

1. Definition of operating system

1.2. Operating system's functions

System's resources management

- System's resources (CPU, memory, IO devices, files...) are **utilized by program** to perform a determined task.
- Programs require resources: **time (CPU-usage)** and **space (memory)**
- The OS has to **manage** the **resource** so that the computer can work in the **most effective** way.

Chapter 1. Operating System Overview

1. Definition of operating system

1.2. Operating system's functions

System's resources management (cont.)

- The OS has to manage the resource so that the computer can work in the most effective way.
 - Provide resource for program **when it's necessary**.
 - **Handle competition**
 - **Decide** the order of **resource providing** for the programs' requirements
- Example: memory resource management (limited)
 - Many program can operate at the same time.
 - Avoid illegal access
 - Data protection (memory sharing: file)

Chapter 1. Operating System Overview

1. Definition of operating system

How do operating system work

▶ ClickView

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Chapter 1. Operating System Overview

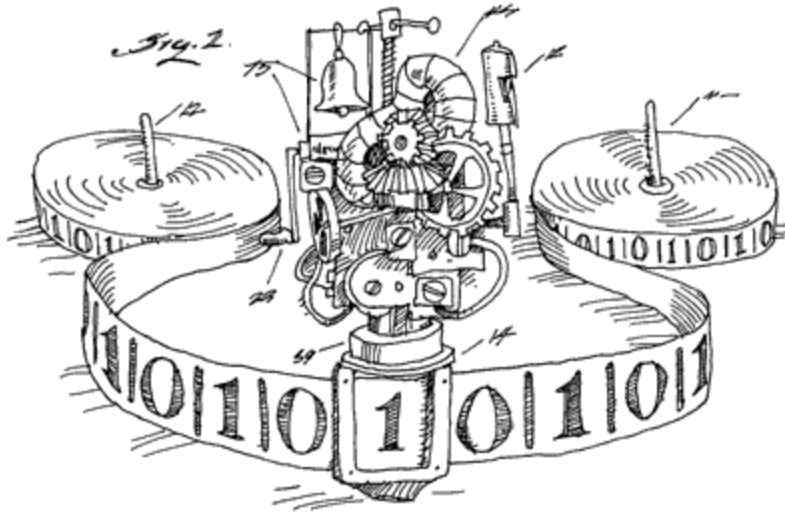
2. History of Operating Systems

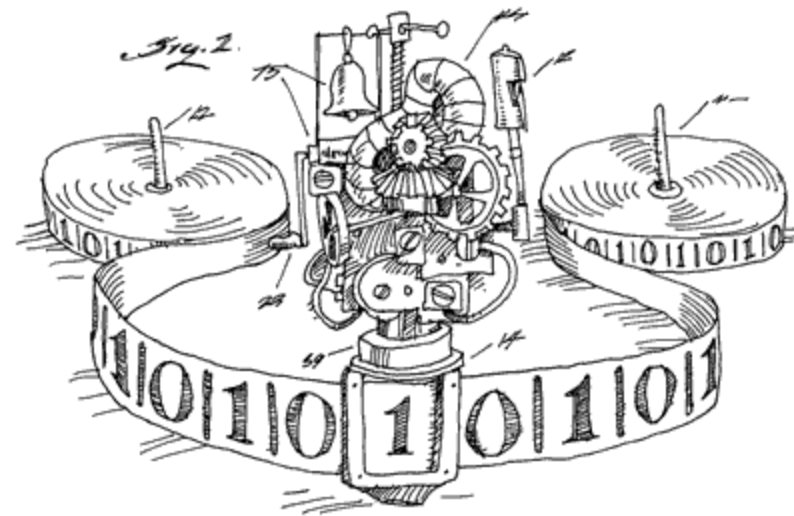
Operating systems development's history

- History of electronic computer
- Operating systems development's history

2.1. History of electronic computer

Development history of electronic computers

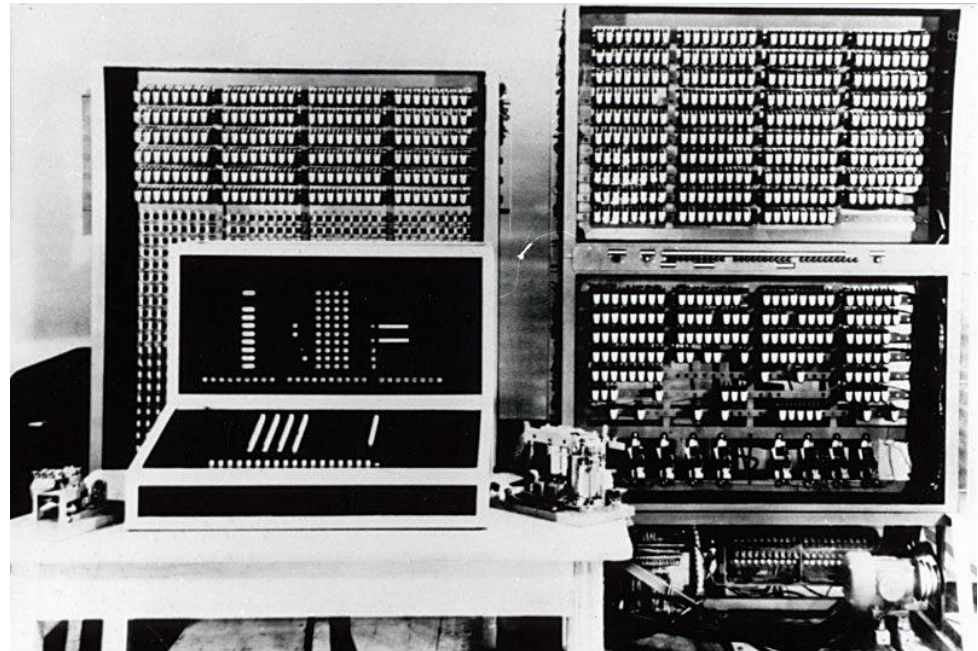
- 1936 - A. Turing & Church present logic computing model and prove the existence of a computer: Turing machine
 - Model of a character processing device; Simple but able to perform all the computer's algorithm
 - A Turing machine that is able to simulate any other Turing machine -> a universal Turing machine
 - *Turing is considered as the father of computer science and artificial intelligence*
- 



Chapter 1. Operating System Overview
2. History of Operating Systems
2.1. History of electronic computer

Development history of electronic computers

- 1941- Konrad Zuse (German) Constructed world's first programmable computer; the functional program-controlled Turing-complete Z3
- Z3: use binary system
- *Has separated memory and controller*
- *Mechanical technique*

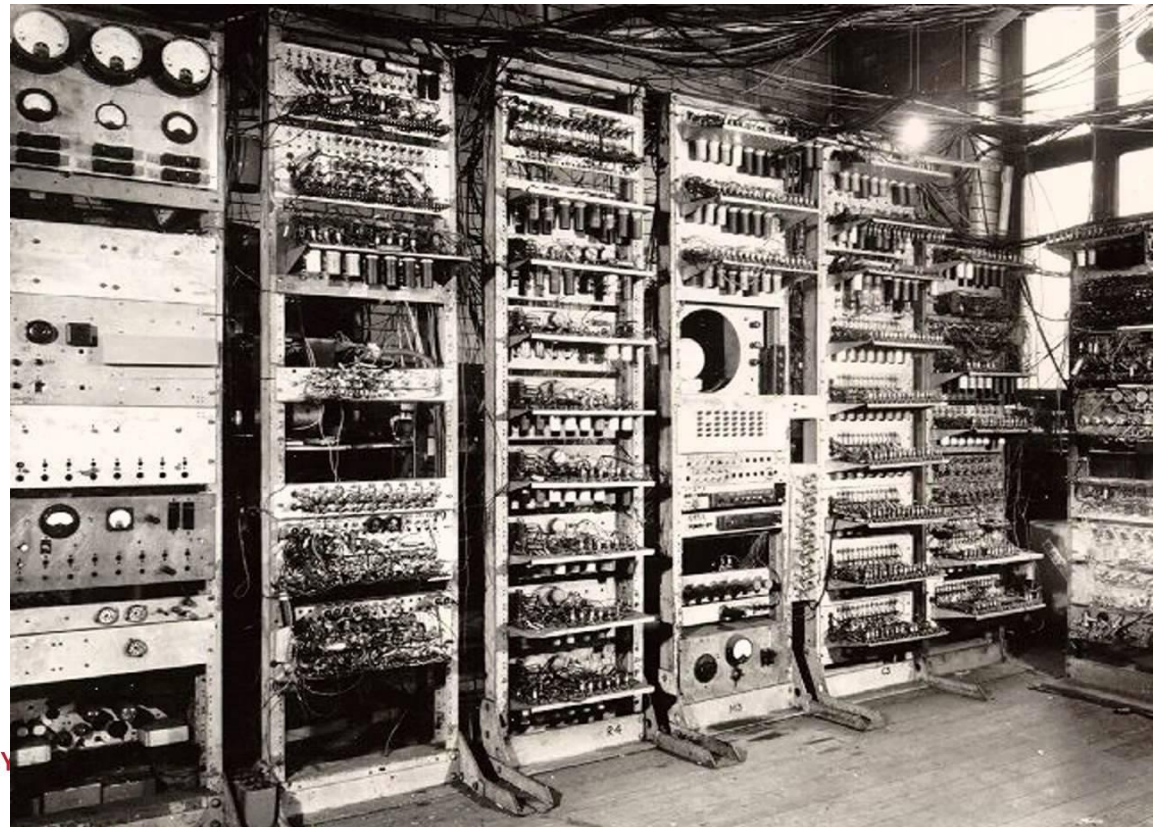


Chapter 1. Operating System Overview
2. History of Operating Systems
2.1. History of electronic computer

Development history of electronic computers

● 1946 ENIAC based on electric bulbs

- 18000 vacuum tube
- 70000 resistance
- 5 million metal connector
- *Faster but least reliable*



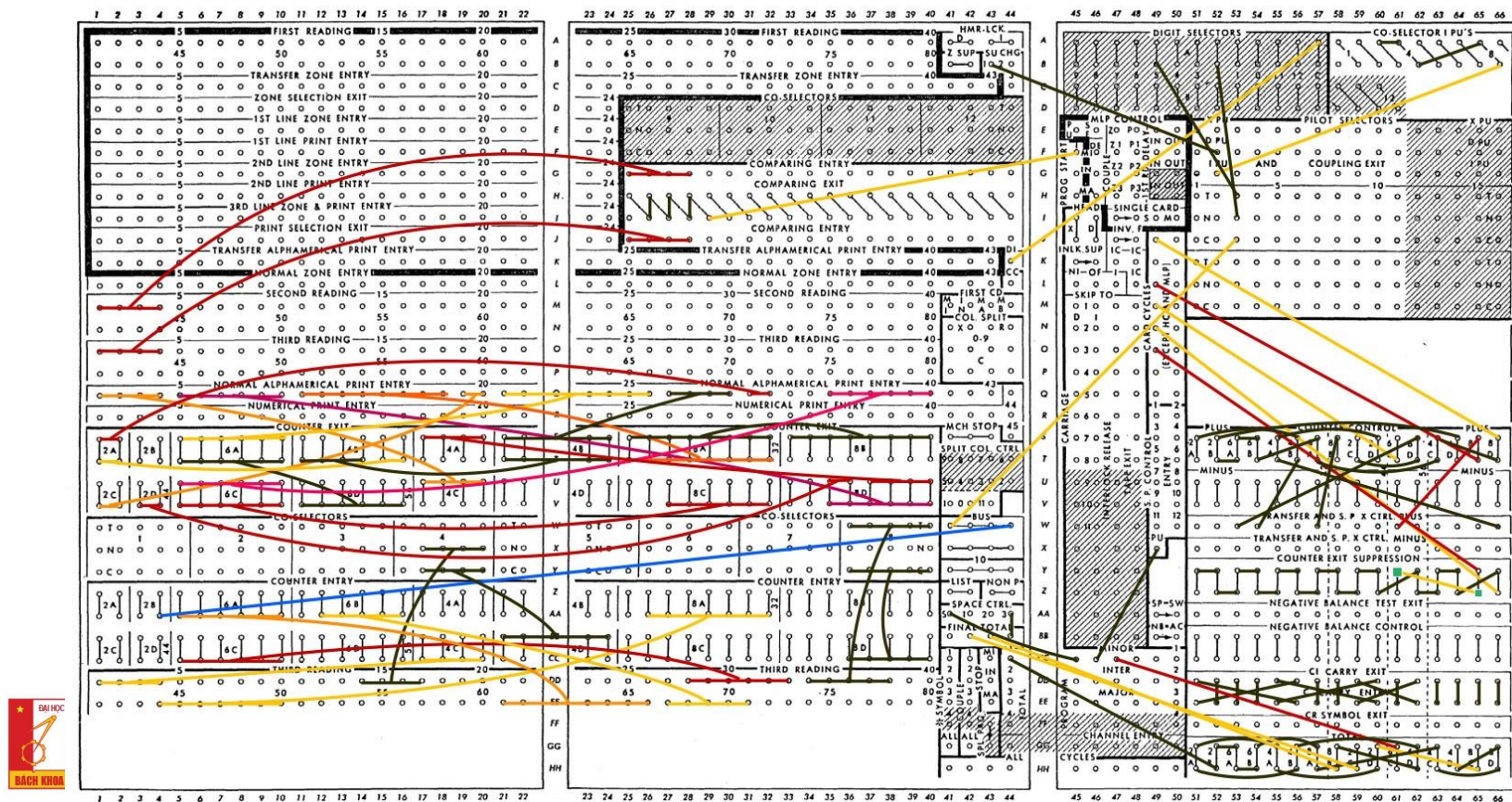
Chapter 1. Operating System Overview

2. History of Operating Systems

2.1. History of electronic computer

Development history of electronic computers

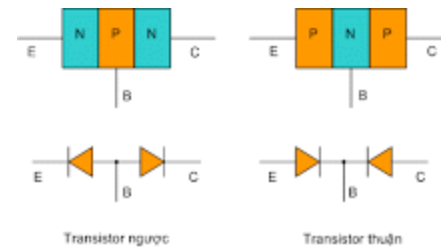
- plug board



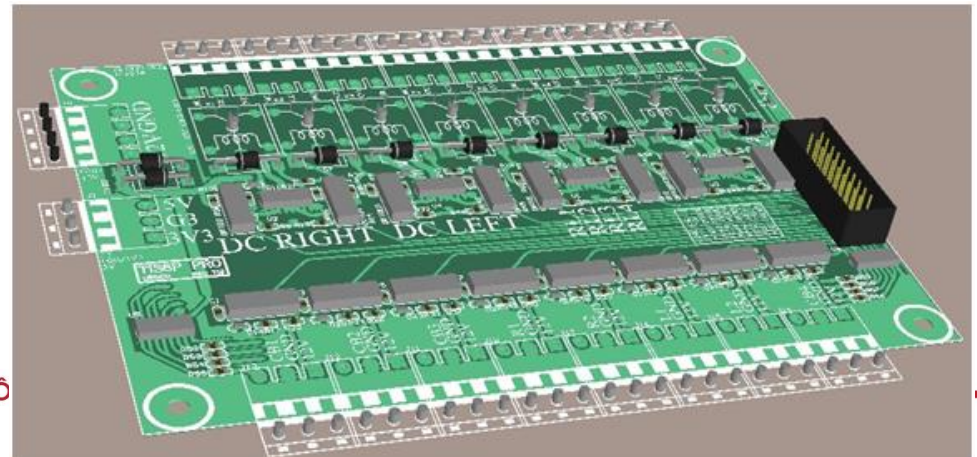
Chapter 1. Operating System Overview
2. History of Operating Systems
2.1. History of electronic computer

Development history of electronic computers

- 1950-1958 Transistor
- 1959-1963 Semiconductor

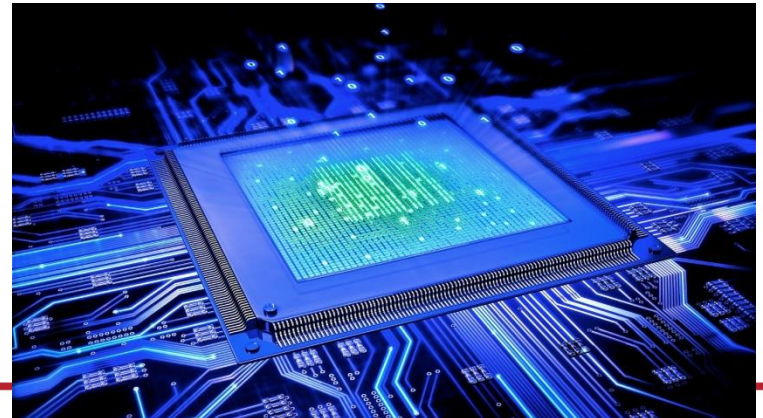


- 1964-1974 Integrated Circuit (IC)



Development history of electronic computers

- 1974-1990 Large scale IC:
Allow CPU, main memory or similar device to be produced in a single integrated circuit
→ new class of smaller, cheaper computer and parallel processor with multi CPUs
- 1990-now Very large scale IC, smart IC



Chapter 1. Operating System Overview

2. History of Operating Systems

Operating systems development's history

- History of electronic computer
- Operating systems development's history

view of change

- 1948-1970 : Hardware expensive; human labor cheap
- 1970-1981 : Hardware cheap; human labor expensive
- 1981- : Hardware very cheap; human labor very expensive
- 1981- : Distributed system
- 1995- : Mobile devices

1948-1970 : Hardware expensive; human labor cheap

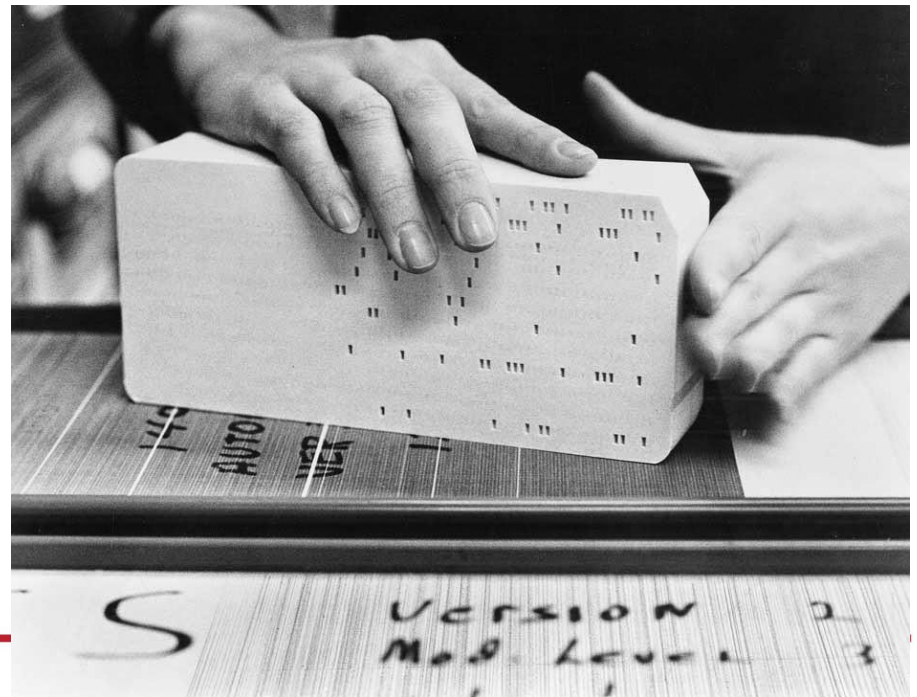
- Computer 1-5 M\$: Nation 's property, mainly used for military's purposes $\Rightarrow$ Require optimization for using hardware effectively
- Lack of human-machine interact
- User, programmer; operator are same group of people
- 1 user at a single time

Chapter 1. Operating System Overview
2. History of Operating Systems
2.2. History of Operating Systems

1948-1970 : Hardware expensive; human labor cheap

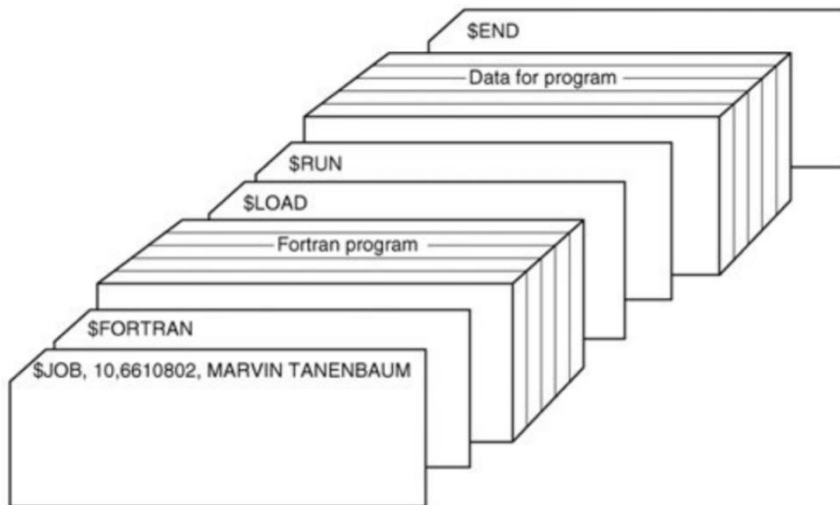
- User wrote program on punched cards
- First card is bootstraps loader is loaded into memory and executed
- Instructions in bootstraps loader fetch into memory and execute instructions on other later cards (application program)
- Check light bulb for results, debug

- Debugging is difficult
- Waste processor time
- Solution: batch processing



Batch processing and professional operator

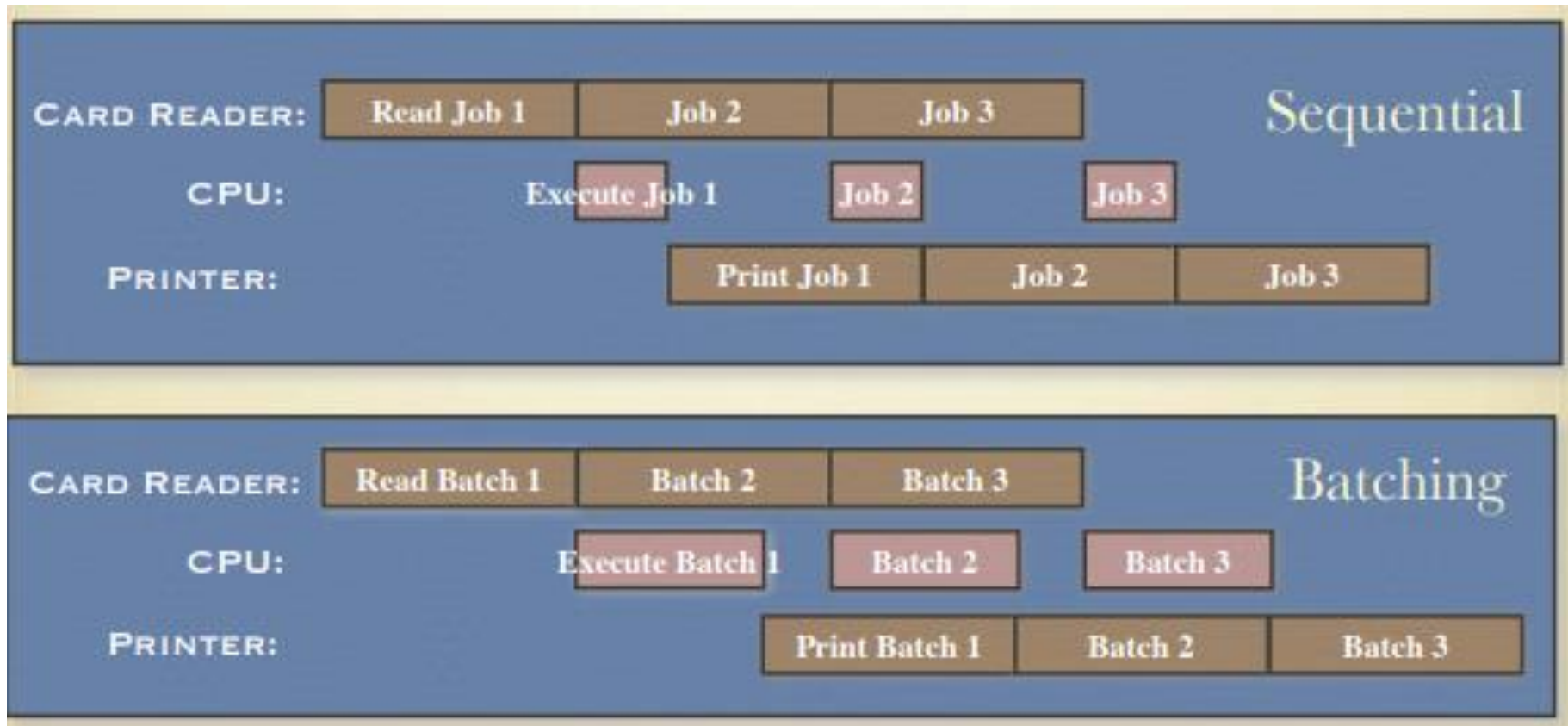
- Programmer give program to operator
- Operator groups program into a single pack (batch)
- Computer reads and run each program consequently
- Operator takes the result, prints out and gives to programmer



- Reduce waiting time between jobs
 - Input/output problem
 - Computer getting faster
 - Card reader still slow
- ⇒ CPU must wait for card reading/writing

Chapter 1. Operating System Overview
2. History of Operating Systems
2.2. History of Operating Systems

Batch and sequential processing



The first Operating System

GM-NAA I/O

General Motors and North American Aviation Input Output System.



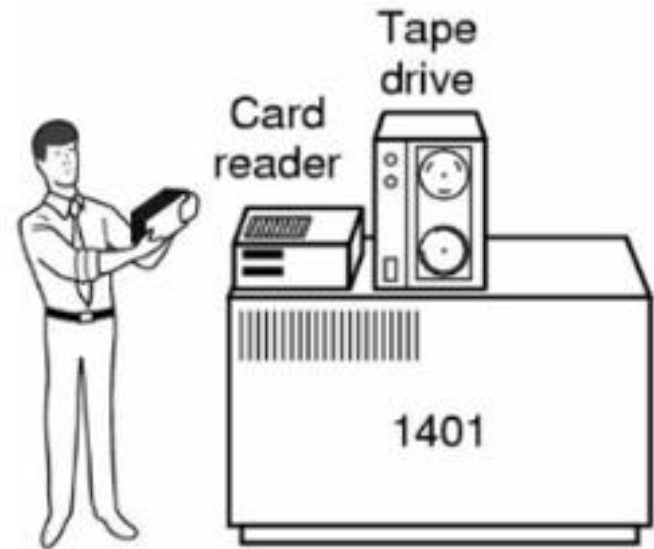
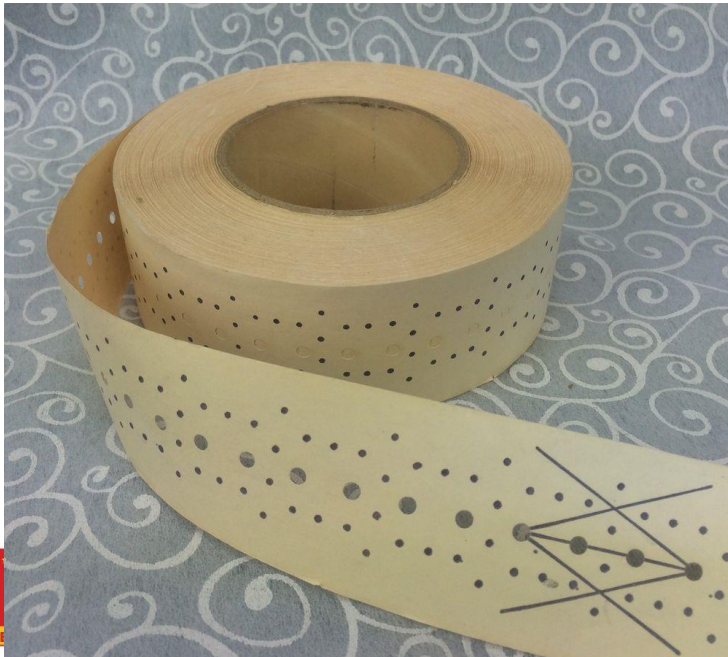
- Created in 1956 by Robert L. Patrick of Owen Mock of North American Aviation.
- Main function: automatically execute a new program once the one that was being executed had finished (batch processing).
- Formed of shared routines to the programs that provided common access to the input/output devices.
- Installed on 40 IBM 704 computers

Chapter 1. Operating System Overview
2. History of Operating Systems
2.2. History of Operating Systems

Independent computer for reading/writing data

- Replace card reader by tape $\Rightarrow$ Independent external computers for reading/writing data

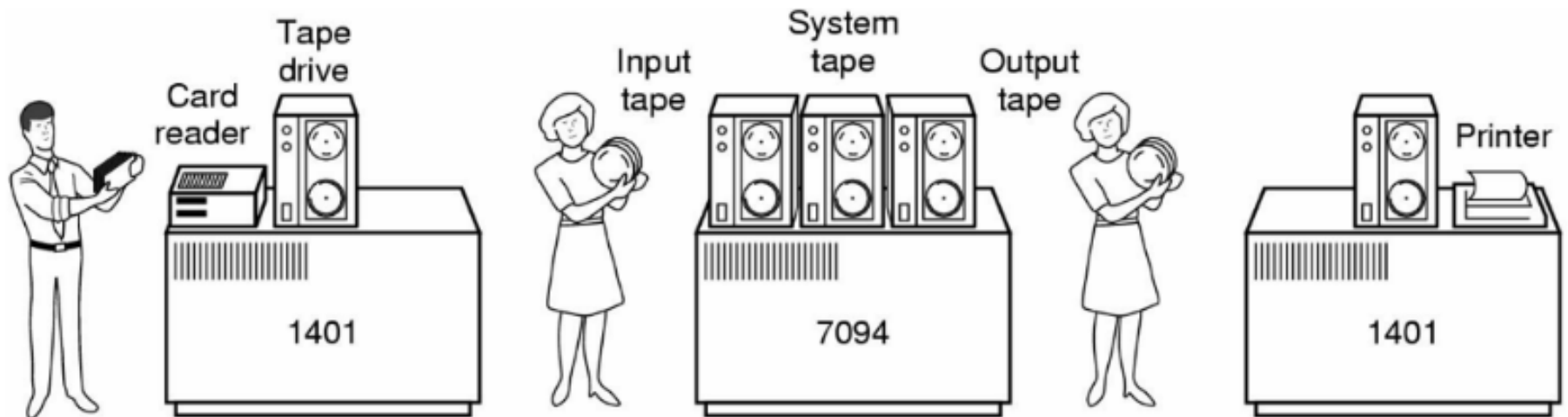
paper tape



Chapter 1. Operating System Overview
2. History of Operating Systems
2.2. History of Operating Systems

Independent computer for reading/writing data

- Replace card reader by tape $\Rightarrow$ Independent external computers for reading/writing data

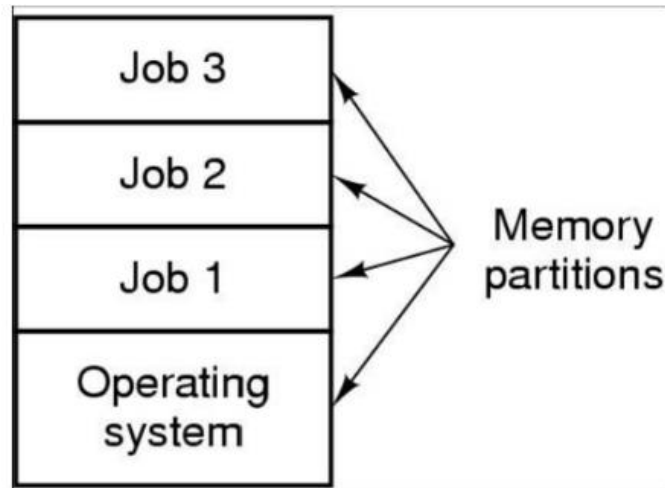


Independent computer for reading/writing data

- External devices are designed to be able to Direct Memory Access, using interrupt and i/o channel
 - OS request I/O device then continue its work
 - OS receive interrupt signal when I/O devices finishes
- ⇒ Allow overlap between computing and I/O
- CPU is reprogrammed to be switch easily between programs

Multi programming

- Hardware: memory space larger and cheaper => Some programs can run simultaneously -> multi programming



- More overlap between computing and I/O
- Require memory protection between programs and keep one crashed program from damaging the system
- Problem: OS must manage all interactions $\Rightarrow$ out of control (OS360: 1000 errors)

1970-1981 :

- Computers prices about 10.000\$ $\Rightarrow$ used widely for different jobs
- OS technology became stable.
- Using cheap terminal device (1000\$) allow many user to interact with the system at the same time

Chapter 1. Operating System Overview
2. History of Operating Systems
2.2. History of Operating Systems

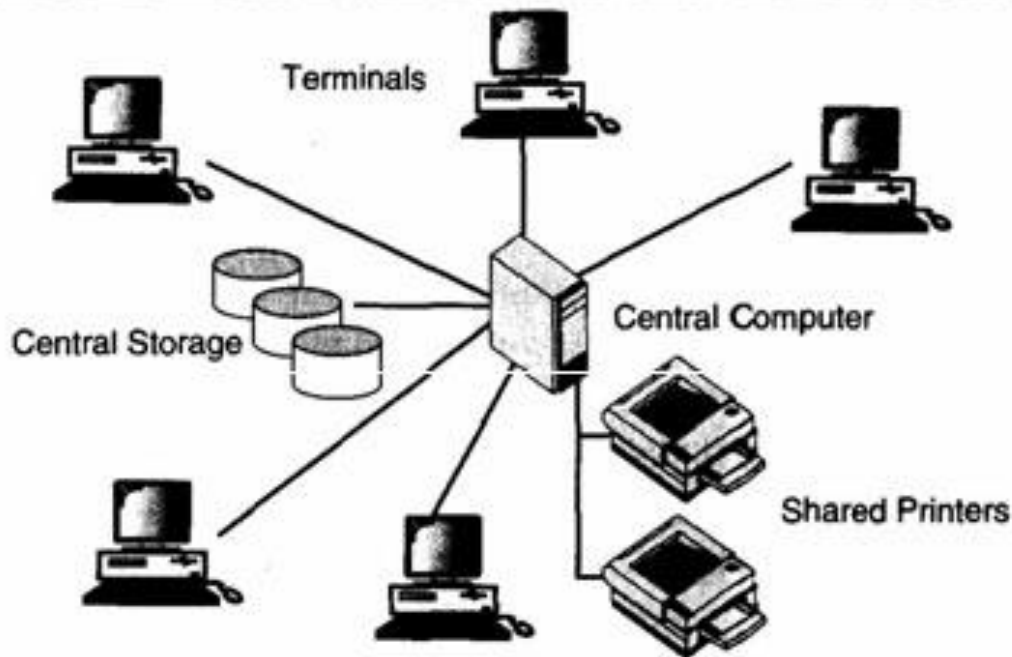
1970-1981 :

- video display terminal
- Ex: DEC VT100 (1978)



Chapter 1. Operating System Overview
2. History of Operating Systems
2.2. History of Operating Systems

Time sharing system



Time-sharing Environment

1970-1981 :

- User perform different works (text editor, chat, program debugging,..) $\Rightarrow$ require system to be exploited effectively
- Example: a PC: 10M calculation/s; typing speed 0.2s/1 character $\Rightarrow$ lost 2M calculation per one typing
 $\Rightarrow$ Time sharing operating system
 - Problem: system's response time
- Computer network was born (ARPANet : 1968) Communication between computer; Protection against network attack

1981-1995 :

- Computers prices about 1000\$;
- Human labor 100K \$/year

If you can make someone 1% more efficient by giving them a computer, it's worth it!

⇒ Computers are used more widely for working more effectively

1981-1995 :

- Personal computing
 - Cheap computer, single person can afford (PC).
 - OS on PC
 - Hardware resources are limited (Early : 1980s)
 - OS become library of available procedures
 - Run 1 program at a time (DOS)
 - PC become more powerful
 - OS meet complex problems: multi tasking, memory protection... (WINXP)
 - Graphical user interface (MAC, WIN,..)
-

Chapter 1. Operating System Overview

2. History of Operating Systems

2.2. History of Operating Systems

- DOS User interface

```
Welcome to FreeDOS

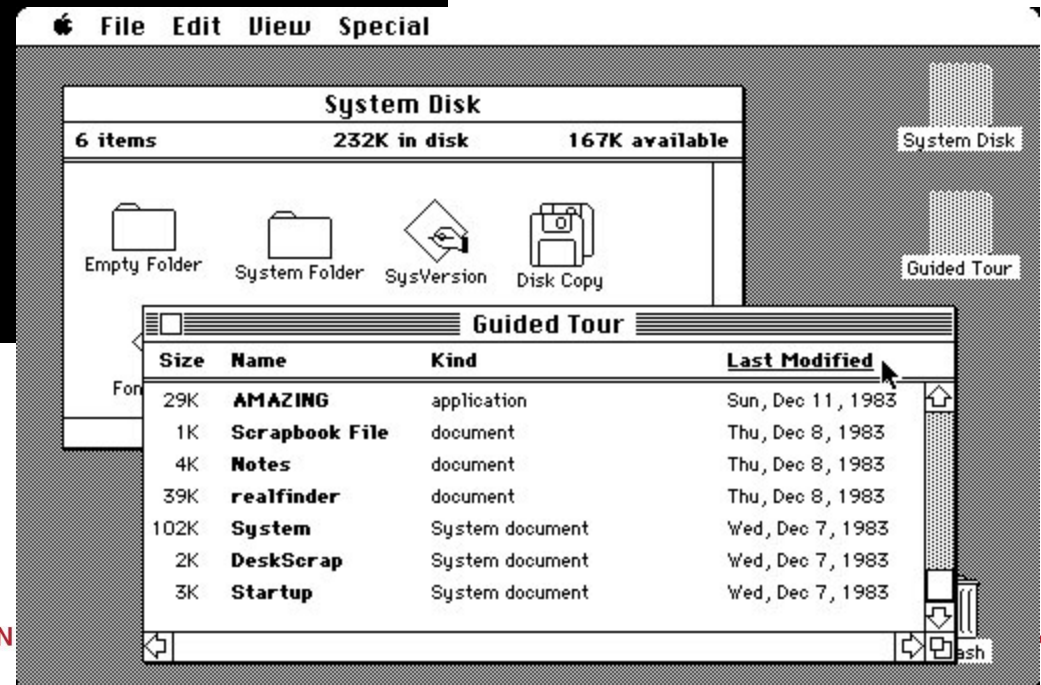
CuteMouse v1.9.1 alpha 1 [FreeDOS]
Installed at PS/2 port
C:\>ver

FreeCom version 0.82 pl 3 XMS_Swap [Dec 10 2003 06:49:21]

C:\>dir
Volume in drive C is FREEDOS_C95
Volume Serial Number is 0E4F-19EB
Directory of C:\

FDOS          <DIR>    08-26-04   6:23p
AUTOEXEC.BAT      435    08-26-04   6:24p
BOOTSECT.BIN      512    08-26-04   6:23p
COMMAND.COM     93,963    08-26-04   6:24p
CONFIG.SYS        801    08-26-04   6:24p
FDSBOOT.BIN       512    08-26-04   6:24p
KERNEL.SYS      45,815    04-17-04   9:19p
6 file(s)         142,038 bytes
1 dir(s)      1,064,517,632 bytes free

C:\>_
```

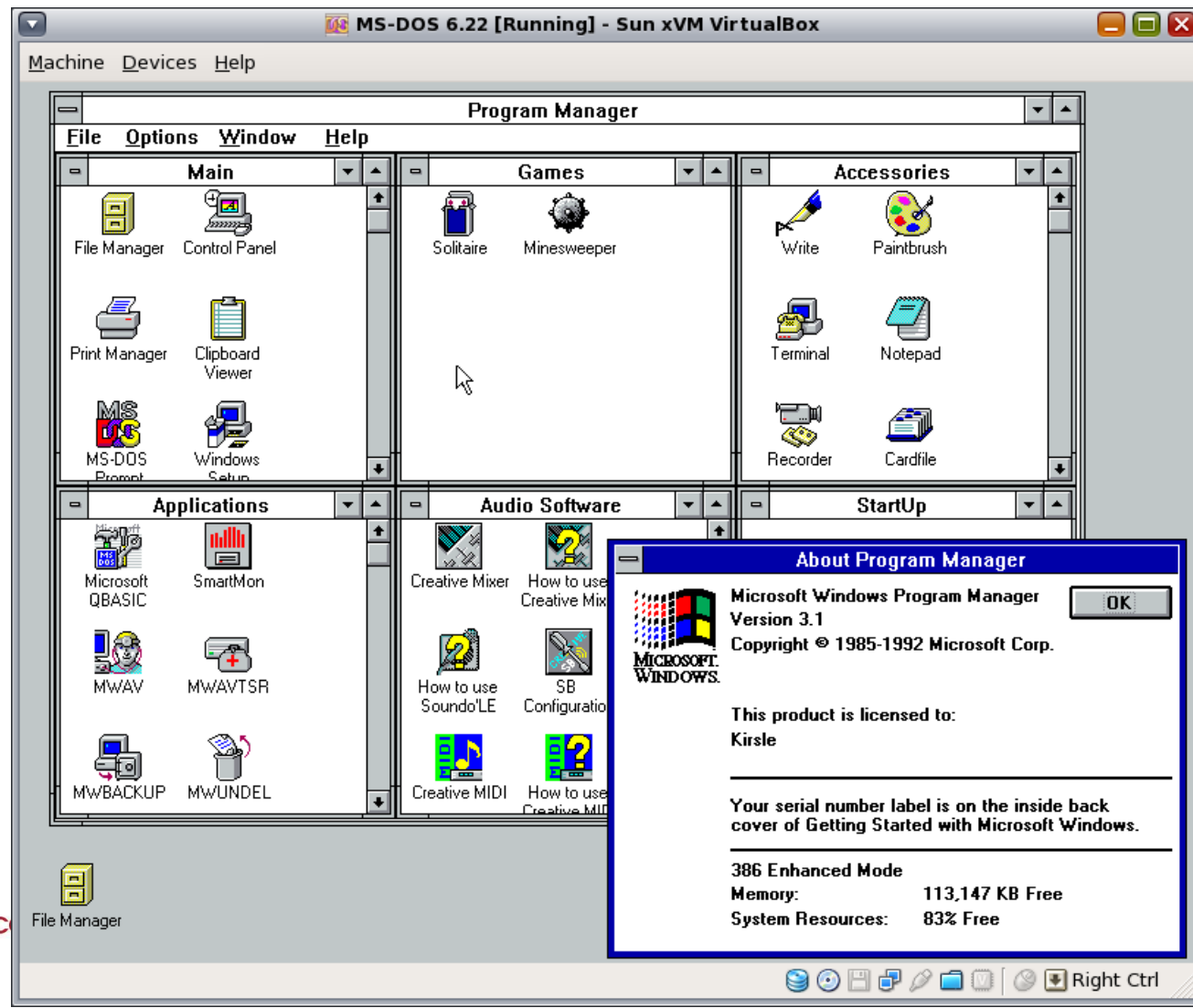


- * Macintosh User interface

Chapter 1. Operating System Overview

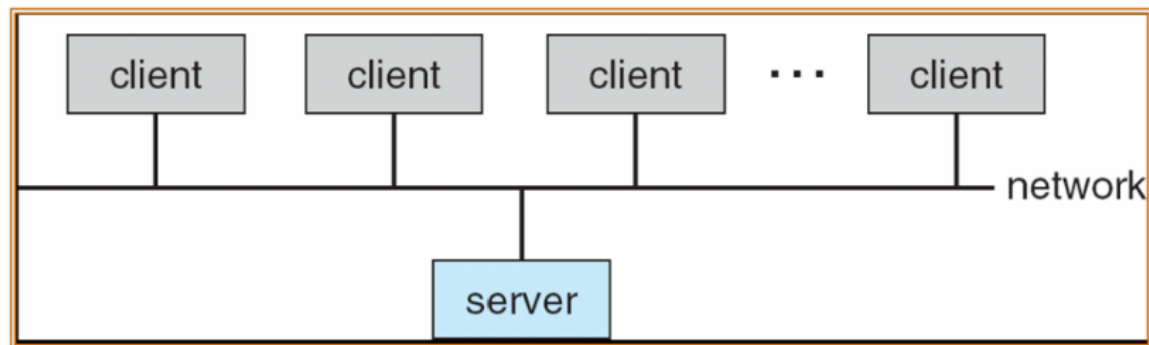
2. History of Operating Systems

2.2. History of Operating Systems



Distributed systems

- Development time of networking and distributed operating systems



- Local area network
 - Computers share resources: printer, File servers,...
 - Client / Server model
- Services
 - Computing, storage
 - Services provided through Internet.

Distributed systems

- Problems
 - Transmission delay; bandwidth, reliable...
 - Virus (love letter virus 05/2000),...
- >45 millions computers were infected
- Stole information
- Auto send emails from contact list
- Download Trojan



Mobile devices

Mobile devices become more popular

- Phone, Laptop, PDA . . .
- Small, changeable and cheap → More computers/human
- Limited ability: speed, memory,...



Mobile devices

- Wide area network, wireless network
 - Traditional computer divided into many components (wireless keyboards, mouse, remoting storage)
- peer-to-peer system
 - Devices with the same role working together
 - “Operating system’s” components are spread globally
- Cloud computing
 - Cloud operating system
 - Ex: Microsoft Windows Azure, Google Chrome OS

Conclusion

- The development of the operating systems are strongly connected with the computers' development
- Operating systems have to change with changing technology

Chapter 1 Operating System Overview

- ① Operating system Definition
- ② History of Operating Systems
- ③ **Definition and Classifications**
- ④ Definitions in operating systems
- ⑤ Operating System structures
- ⑥ Basic Properties of Operating Systems
- ⑦ Principles of Operating Systems

Chapter 1. Operating System Overview

3. Operating System definition and classification

Operating System definition and classification

- Definitions
- Classifications

Chapter 1. Operating System Overview

3. Operating System definition and classification

3.1. Definition of operating system

Observer's perspective

- Different objects have different requirements for OS
- Different observing perspectives $\Rightarrow$ different definitions

❖ User

❖ Manager

❖ Technical perspective

❖ System engineer perspective

Chapter 1. Operating System Overview

3. Operating System definition and classification

3.1. Definition of operating system

User's perspective:

“A system of programs that help exploit the computing system conveniently”



Chapter 1. Operating System Overview

3. Operating System definition and classification

3.1. Definition of operating system

Manager's perspective

“A system of programs that help manage the computing system's resources effectively”



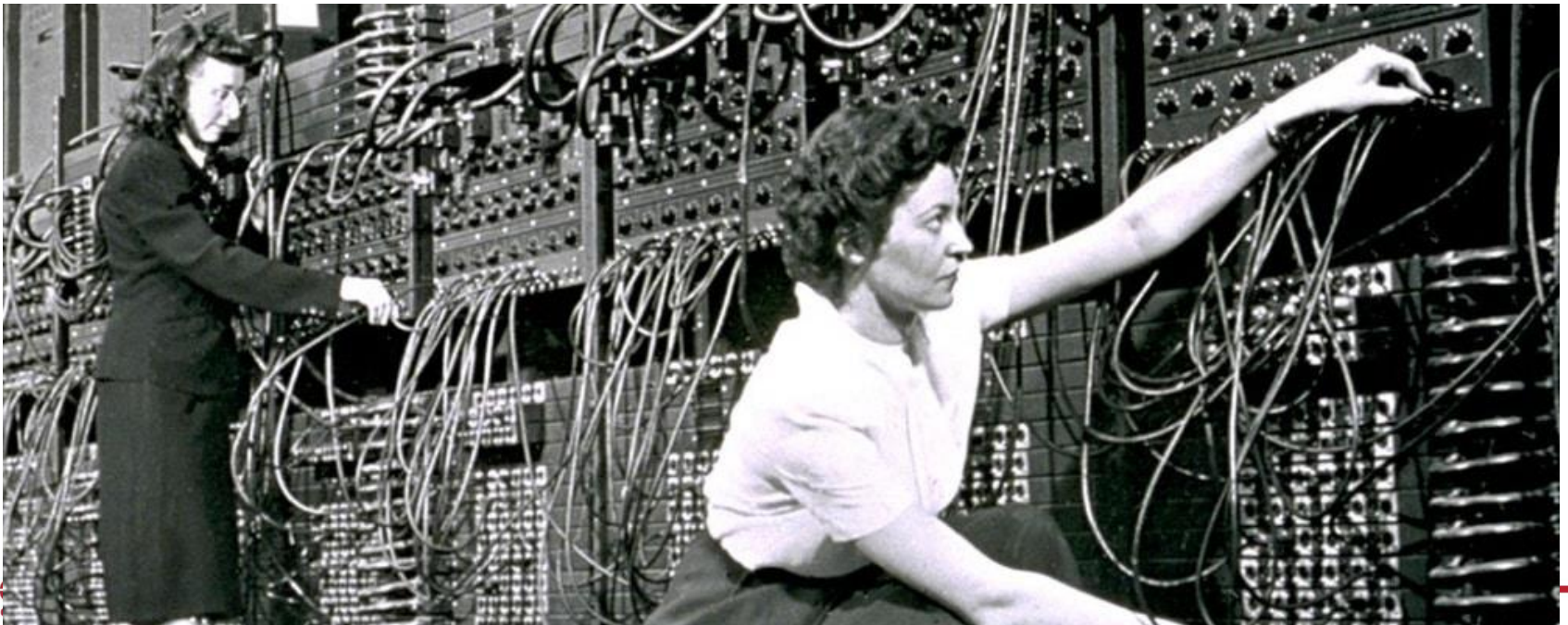
Chapter 1. Operating System Overview

3. Operating System definition and classification

3.1. Definition of operating system

Technician's perspective

“A system of programs equipped for a specific computer to make a new logic computer with new resources and new abilities”



Chapter 1. Operating System Overview

3. Operating System definition and classification

3.1. Definition of operating system

System engineer's perspective

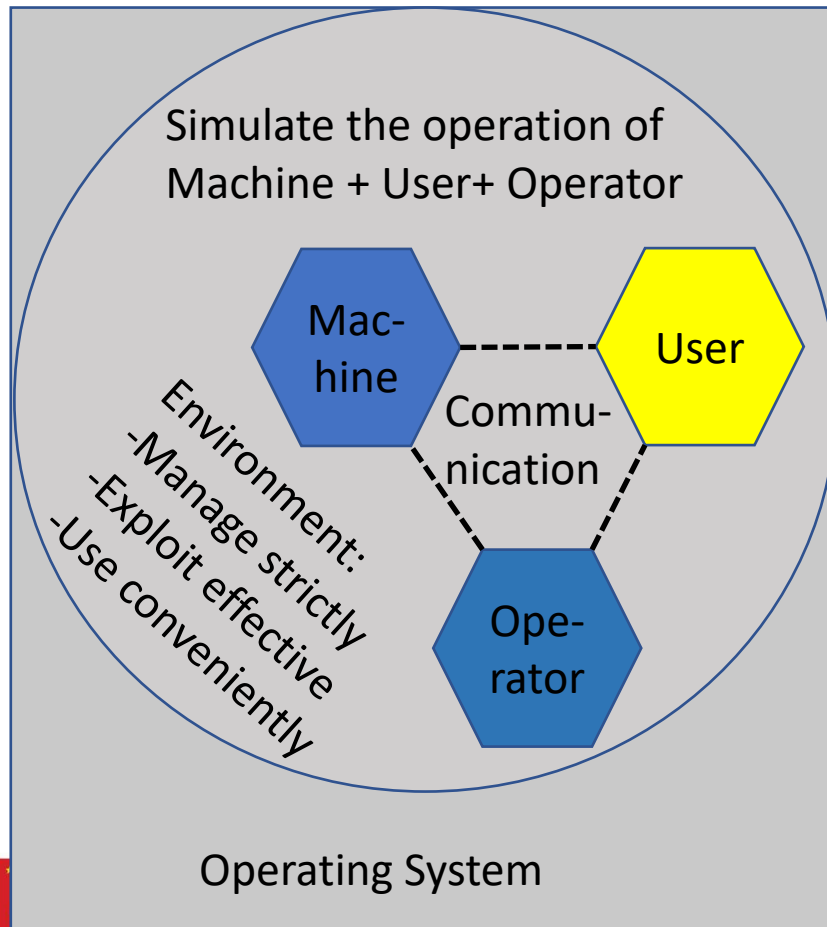
“A system of programs that model and simulate the operation of the computer, user, and operators. It works in a communicating mode to make a convenient environment for exploiting the computer system and maximum resource management.”

Chapter 1. Operating System Overview

3. Operating System definition and classification

3.1. Definition of operating system

System engineer's perspective



- Simulate 3 roles $\Rightarrow$ require 3 types of languages
- **Machine language**
 - The only working language of the system
 - All other languages have to be translated into machine language
- **System operation's language**
 - OS commands (DOS: Dir, Del.; Unix: ls, rm,..)
 - Translated by the Shell
- **Algorithm language**
 - Programming language
 - Compiler

Chapter 1. Operating System Overview

3. Operating System definition and classification

Operating System definition and classification

- Definitions
- Classifications

Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

- Batch processing single program system
- Batch processing multi program system
- Time sharing system
- Parallel system
- Distributed system
- Real-time processing system

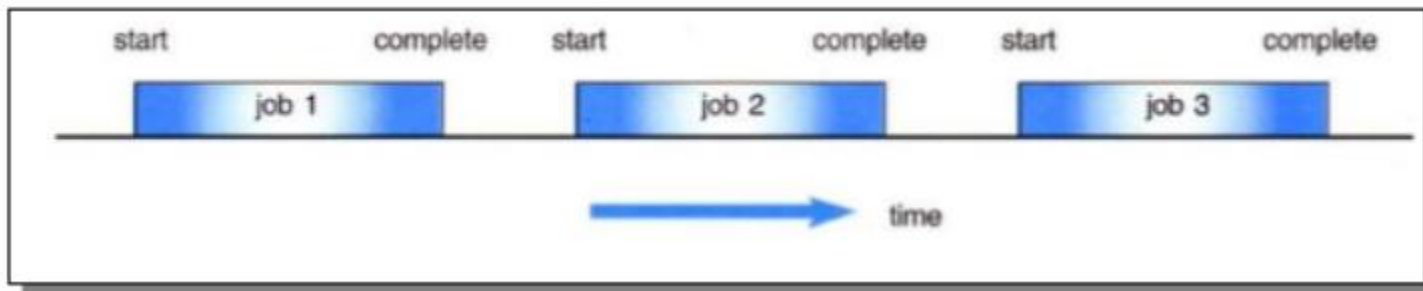
Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Batch processing single program system

- Programs are performed **consequently** follow **predetermined instructions**
- When a program finished, the system **automatically run** the **next program** without any external intervention



- Require a **special process to supervise** the sequence of jobs and the supervisor must stay **permanent in the memory**
- Need to **organize a job queue**

Chapter 1. Operating System Overview

3. Operating System definition and classification

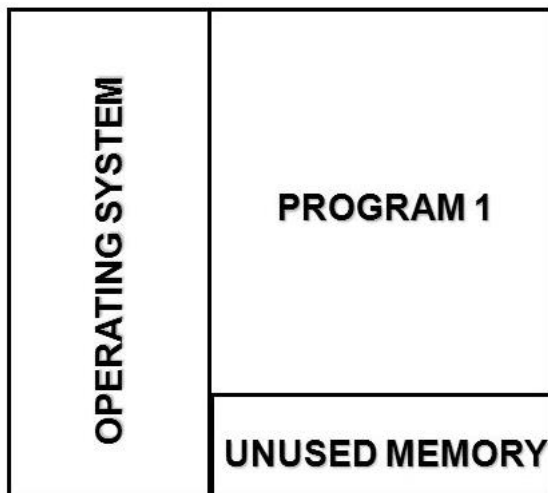
3.2. Operating System Classification

Batch processing multi-program system

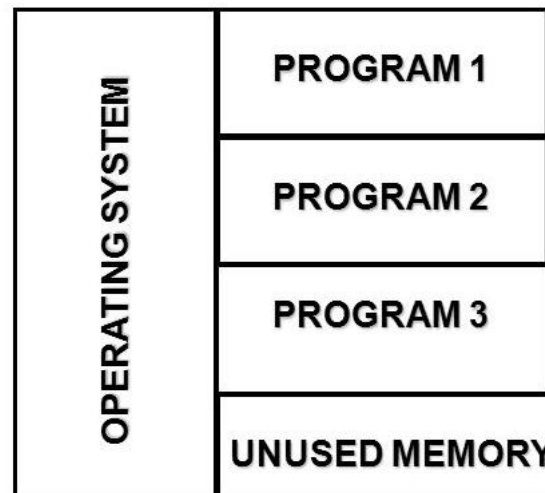
- Problem: when a program access an I/O device, processor has to wait
- Solution: allow many programs to run at the same time

MULTIPROGRAMMING

TRADITIONAL SINGLE-PROGRAM SYSTEM



MULTIPROGRAMMING ENVIRONMENT



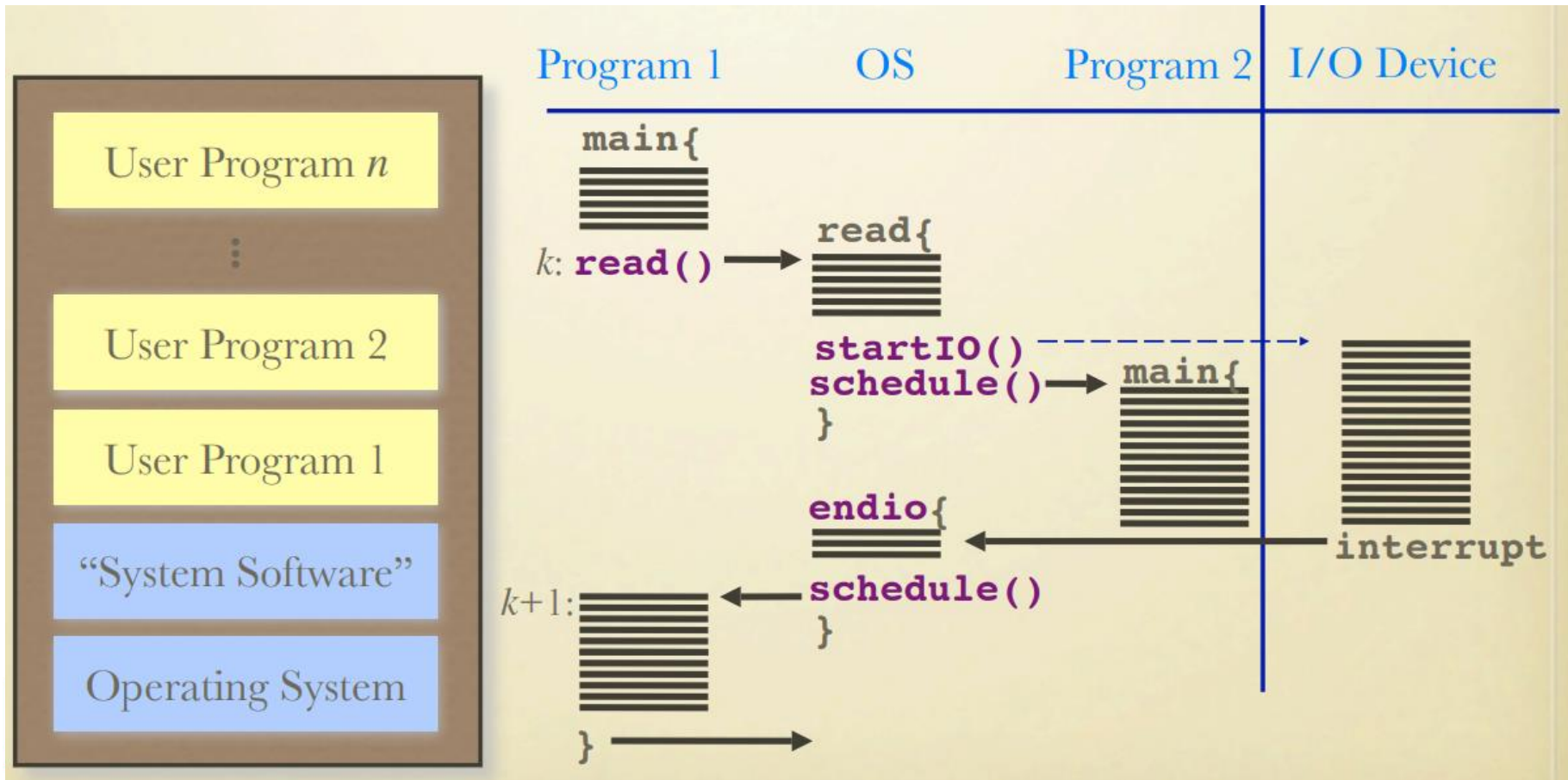
Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Batch processing multi-program system

- Keep several jobs in memory and multiplex CPU between jobs



Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Batch processing multi-program system

- **Save memory** (no need to load all the program into the memory)
- **Reduce processor spare-time**
- High cost for **processor scheduling**. Which program can use processor next?
- How to solve the **memory sharing problem** between programs ?

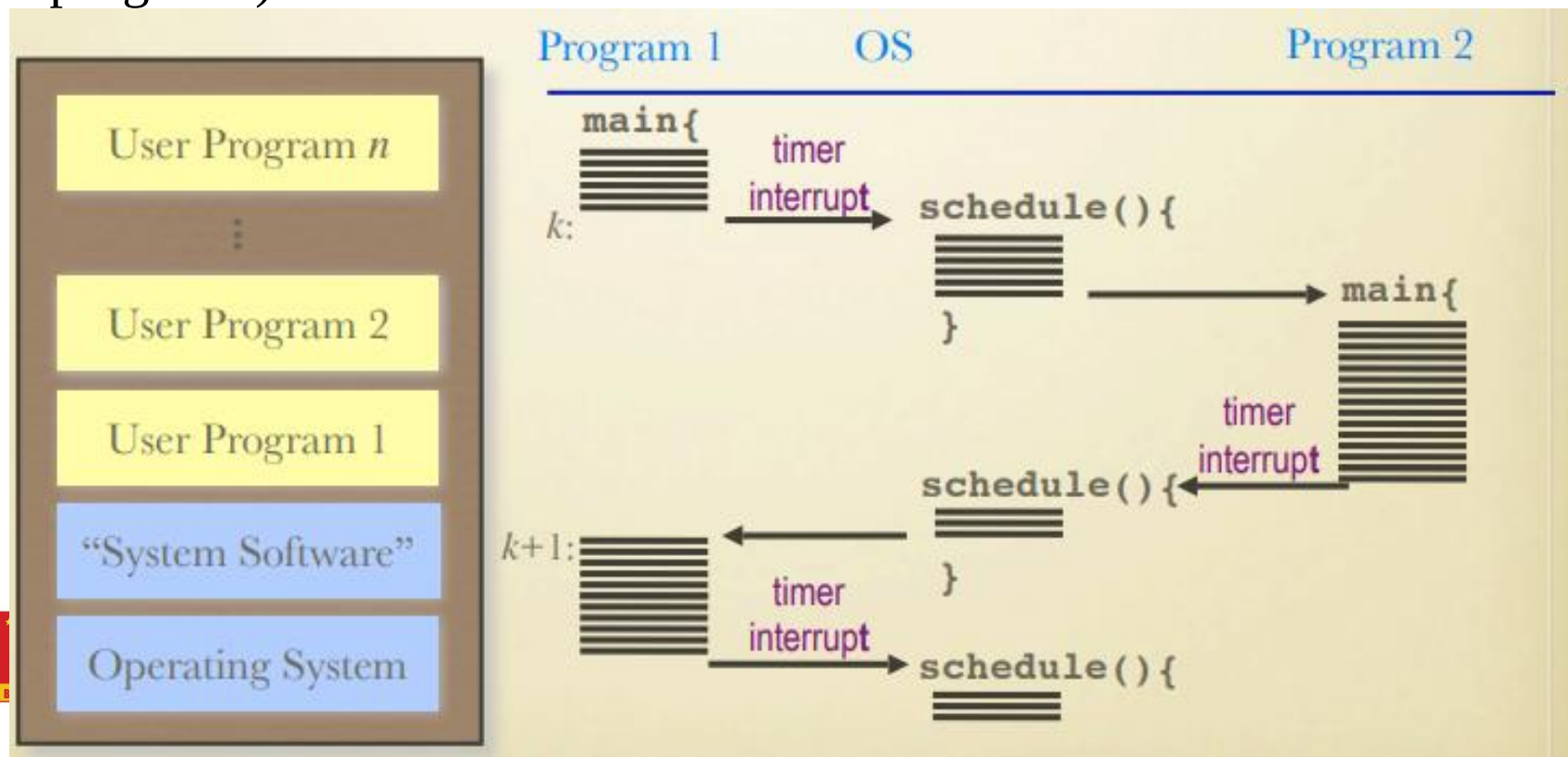
Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Time sharing system

- Processor's usage allowance **time** is **shared** among **ready-to-run programs**
- Similar to batch processing multi program system (only load part of the programs)



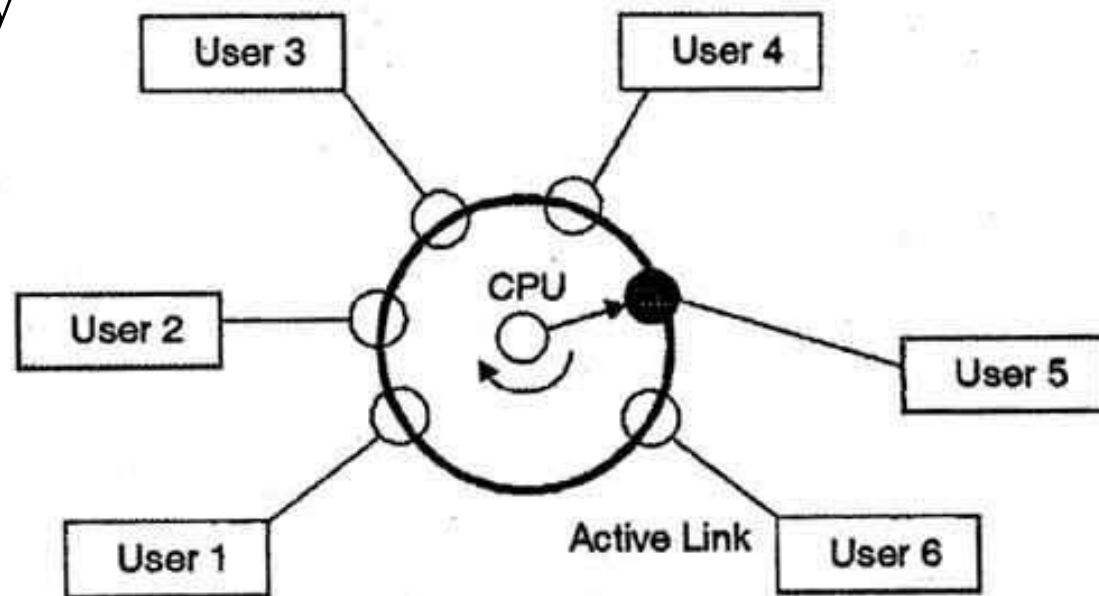
Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Time sharing system

- Processor is **issued** mainly by the **OS** $\Rightarrow$ how ? $\Rightarrow$ **Chapter 2**
- **Swapping time** between programs are **small** $\rightarrow$ programs seem to run **parallel**
- Usually called: **Multi tasking** operating system (Windows, Unix)



Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Parallel system

- Constructed for system that has many processors
 - Many processors, works are done faster
 - More reliable: one processor breaks down will not affect the system
 - Advantage over single processor computer due to memory, peripheral devices sharing...
- Symmetric multi processing (SMP: symmetric)
 - Each processor run a single program
 - Processors communicate via a shared memory
 - Fault tolerance mechanism and optimal load balance
 - Problem: processor synchronization
 - Example: WinNT operating system

Chapter 1. Operating System Overview

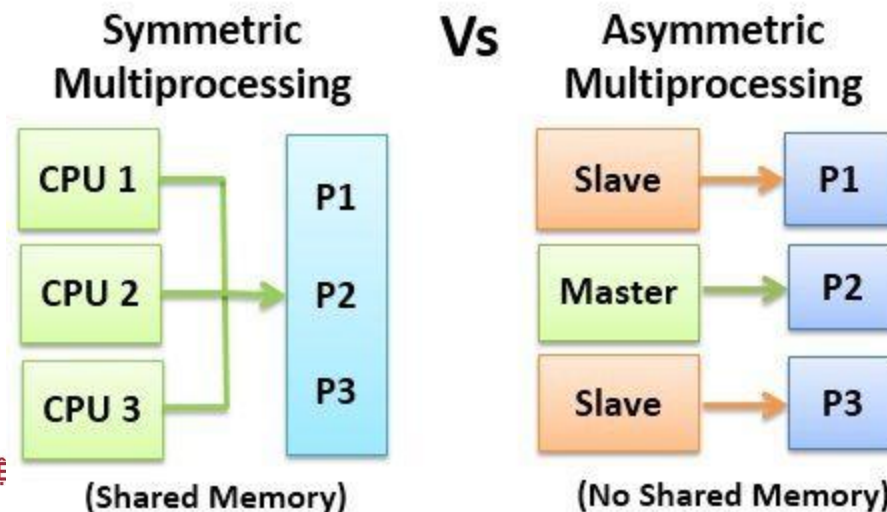
3. Operating System definition and classification

3.2. Operating System Classification

Parallel system

(cont.)

- Asymmetric multi processing (ASMP: asymmetric)
 - One processor **controls** the **whole system**
 - Other processors **follow** the main processor's **commands** or predetermined instructions
 - This model has the **master-slave relation** form: The main process will make schedule for other processors
 - Example: IBM System/360



Chapter 1. Operating System Overview

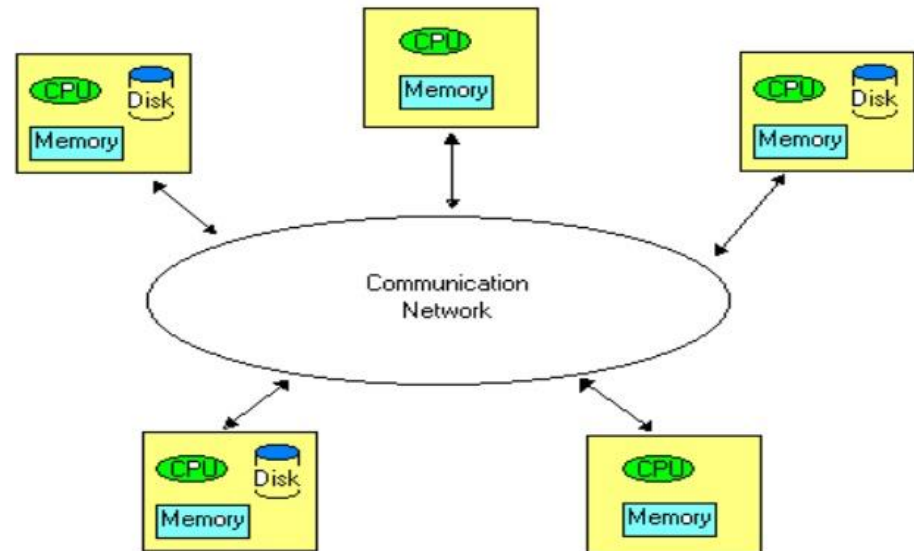
3. Operating System definition and classification

3.2. Operating System Classification

Distributed system

- Each processor has a **local memory** and **communicate** via **transmission lines**
- Processors are **different** from sizes to functions (personal machine, workstation, mini computer,...)

Architecture of Distributed OS



- Ex: Solaris, AIX

Chapter 1. Operating System Overview

3. Operating System definition and classification

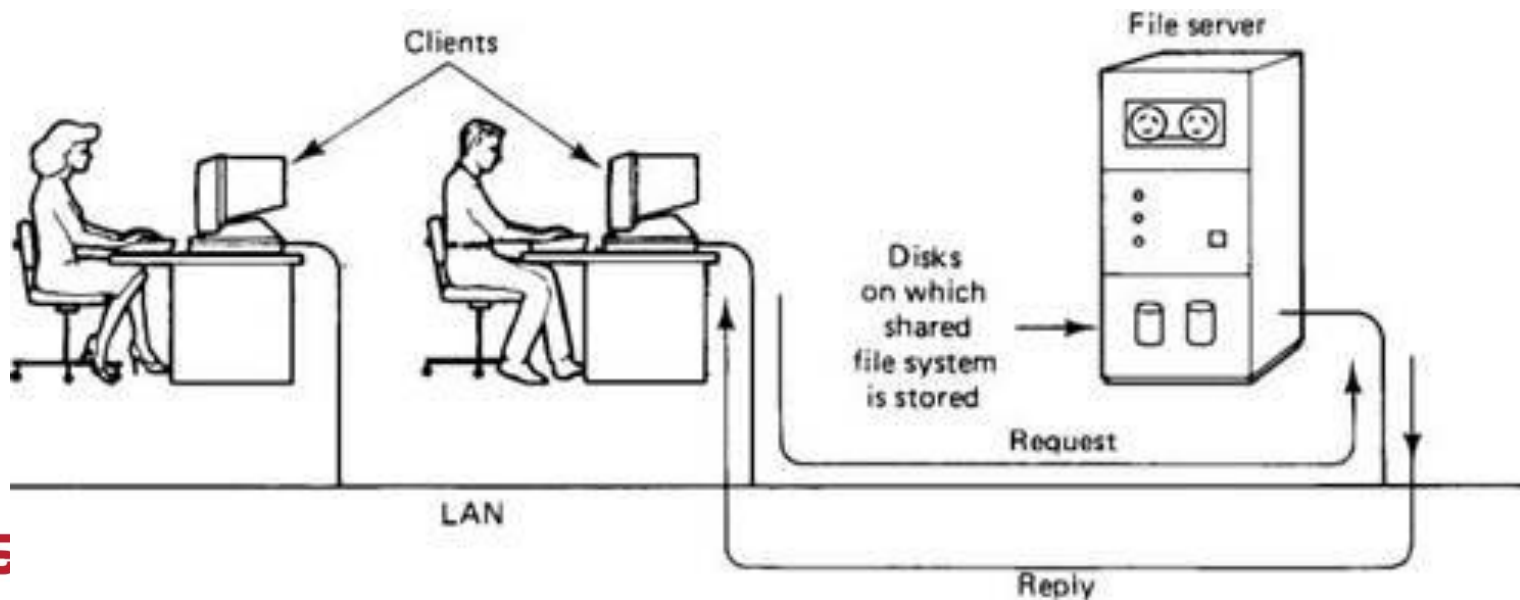
3.2. Operating System Classification

Distributed system

II

Used for

- Resource sharing
- Increase computing speed
- Safety



Chapter 1. Operating System Overview

3. Operating System definition and classification

3.2. Operating System Classification

Real-time processing system

- Used mainly in **controlling** field.
- Solve a problem **no late** than a specific time.
 - Each problem has a **deadline**
 - The system must **generate correct result** in a determined time period
- This OS requires **high cooperation** between **software** and **hardware**.
- Example: VxWorks, RT Linux



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- ① Operating system Definition
- ② History of Operating Systems
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- ④ Concepts in operating systems
- ⑤ Operating System structures
- ⑥ Basic Properties of Operating Systems
- ⑦ Principles of Operating Systems

Chapter 1 Operating System Overview
4. Concepts in operating systems

Concepts in operating systems

- Process
- System's resources
- Shell
- Kernel
- System calls

Chapter 1 Operating System Overview
4. Concepts in operating systems
4.1. Process and Thread

Process

- A running program
 - Codes: Program's executable instruction
 - Program's data
 - Stack, stack pointer, registers
 - Information that is necessary for running program
- Process $\gg$ program
 - Program: a passive object, contains computer's instructions to perform a specific task
 - Process: program's active state.

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.1. Process and Thread

BASIS FOR COMPARISON	PROGRAM	PROCESS
Basic	Program is a set of instruction.	When a program is executed, it is known as process.
Nature	Passive	Active
Lifespan	Longer	Limited
Required resources	Program is stored on disk in some file and does not require any other resources.	Process holds resources such as CPU, memory address, disk, I/O etc.

Multi-process timesharing system

- Periodically: OS pauses 1 process and starts another process
 - Need to store processes' information $\Rightarrow$ process table
- 1 process can start other process
 - Ex: OS's Shell start a process to perform the command; when the command is done, terminate the started process
- Process can exchange information
- 1 process can include many threads

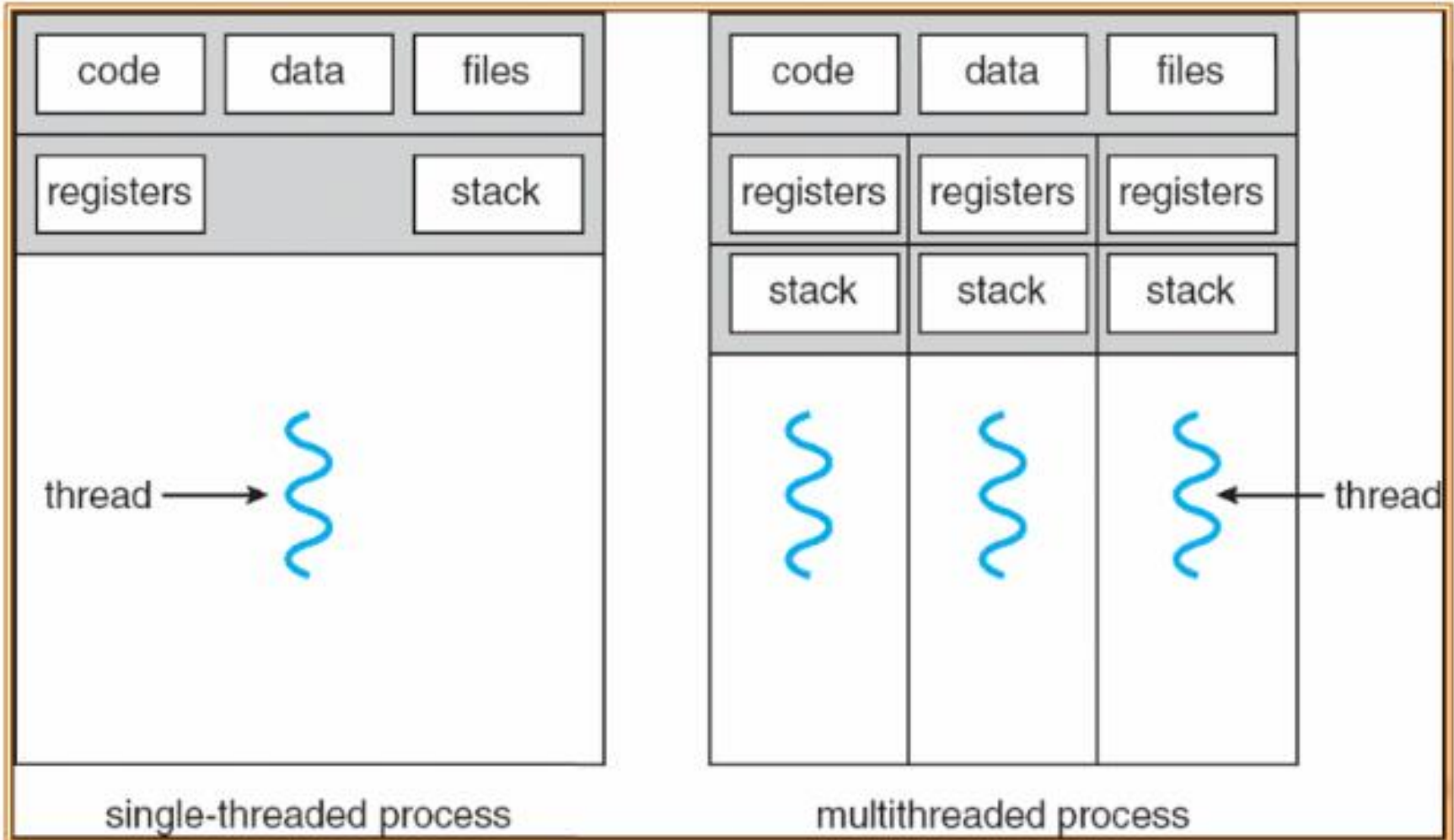
Thread

- A sequence/thread of instructions executed in the program
 - Executable code, data
 - Instruction pointer, stack, registers
 - Heavyweight process: contains 1 thread
 - Lightweight process: contains more than 1 thread
 - Multi_Threading model:
 - Threads running parallel, sharing process's global variables
-

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.1. Process and Thread



Chapter 1 Operating System Overview

4. Concepts in operating systems

Concepts in operating systems

- Process and Thread
- System's resources
- Shell
- Kernel
- System calls

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.2 System's resources

Concept of system resources

- Everything that is necessary for a program to be executed
 - Space: System's storage space
 - Time: Instruction executing time, data accessing time
- System's resources
 - Processor
 - Memory
 - Peripheral devices

Chapter 1 Operating System Overview

4. Concepts in operating systems

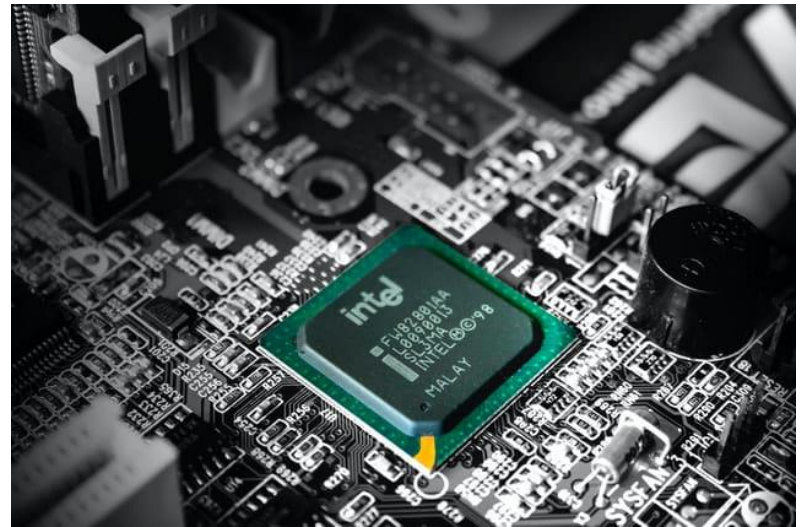
4.2 System's resources

System's resources

(cont.)

Processor

- System's most important component
- Access level: instruction
- Processing time
- Multi-processor system: each processor's time is managed and scheduled independently



Chapter 1 Operating System Overview

4. Concepts in operating systems

4.2 System's resources

Concept

- **Memory**
 - Distinguished by: Storage size, directly access, sequent access
 - Levelled by main memory/internal; extend, external



Chapter 1 Operating System Overview

4. Concepts in operating systems

4.2 System's resources

System's resources (cont.)

- Peripheral devices
 - Retrieve, output information (I/O device)
 - Attached to the system via controller
 - Commonly considered peripheral devices-controller devices



Chapter 1 Operating System Overview

4. Concepts in operating systems

4.2 System's resources

Resource's classification

- **Resource's types**
 - Physical : physical devices
 - Logical: variable; virtual devices
- **Sharing ability**
 - **Sharable**: at a specific time, it can be allocated for different processes. Example: Memory
 - **Non-sharable but dividable**: Processes use the resource follow an order; Example: processor
 - **Non-sharable and non-dividable** : at a specific time, only one process can use the resource. Example: Printer

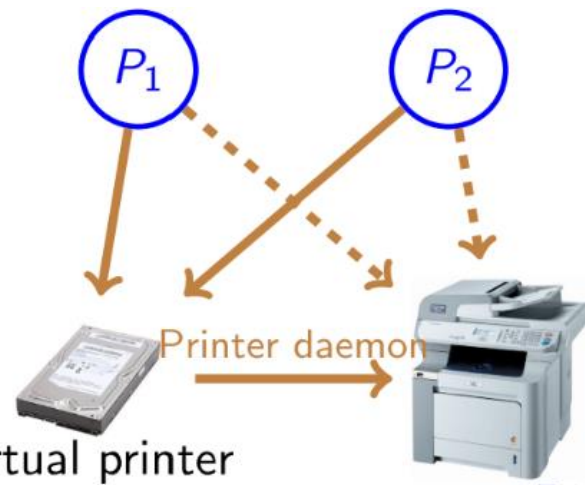
Chapter 1 Operating System Overview

4. Concepts in operating systems

4.2 System's resources

Virtual resource

- provided to a user program in a modified form
- Appears only when the system needs it or when the system creates it
- Automatically disappears when the system terminates or, more precisely, when the process associated with it has terminated.
- Example: Virtual printer



Chapter 1 Operating System Overview

4. Concepts in operating systems

Concepts in operating systems

- Process and Thread
- System's resources
- Shell
- Kernel
- System calls

Chapter 1 Operating System Overview

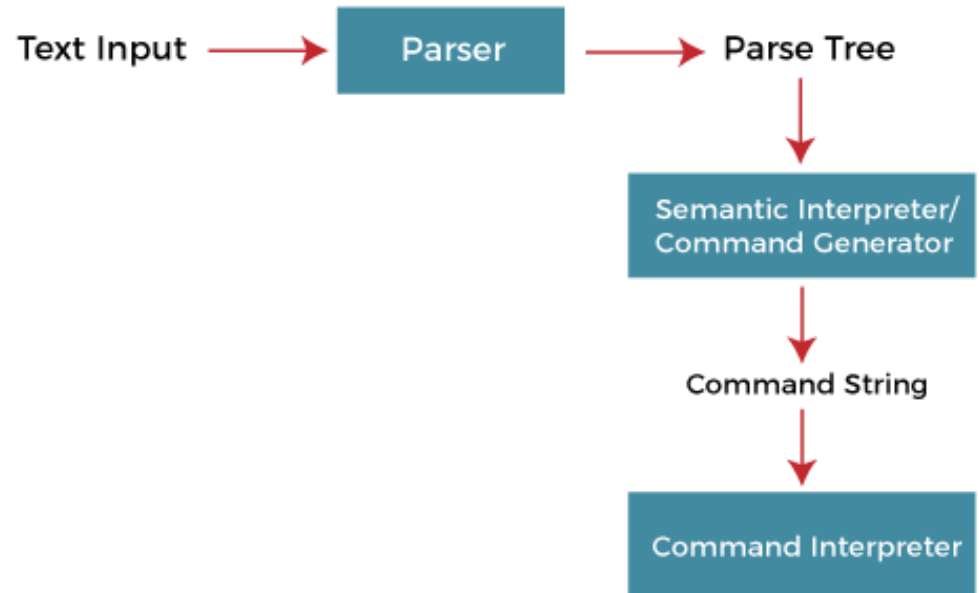
4. Concepts in operating systems

4.3 Shell

- A special process: user and OS communication environment

- Task

- Receive user's command
 - Analysis received command
 - Generate new process to perform command's requirement
- Receives command from command's line or graphical interface

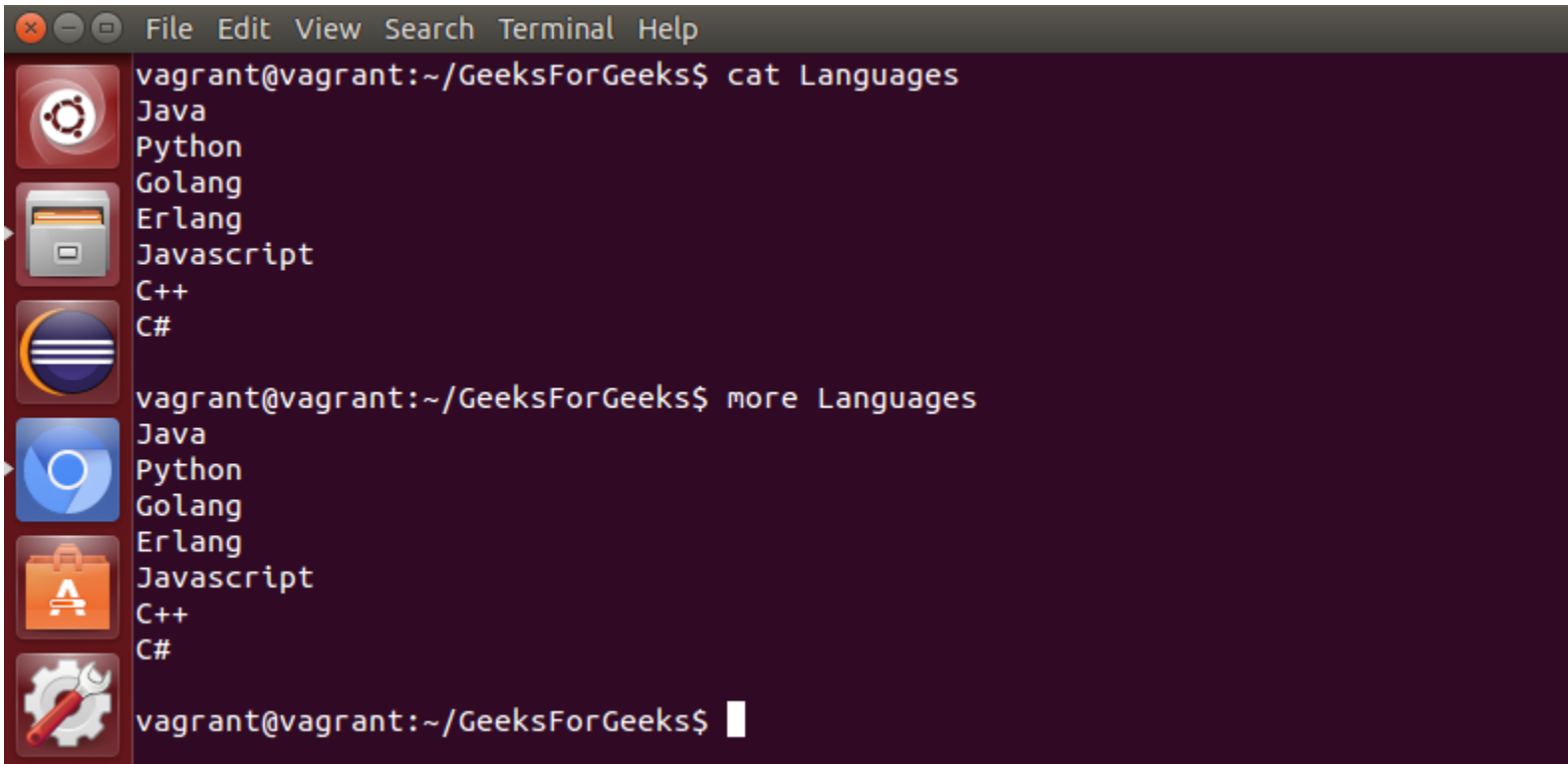


Chapter 1 Operating System Overview

4. Concepts in operating systems

4.3 Shell

- Shell on Linux



The screenshot shows a terminal window with a dark purple background. The window title bar includes 'File Edit View Search Terminal Help'. The prompt is 'vagrant@vagrant:~/GeeksForGeeks\$'. The command 'cat Languages' has been executed, displaying the following text:

```
Java
Python
Golang
Erlang
Javascript
C++
C#
```

The command 'more Languages' has been executed, displaying the same text as above.

The prompt 'vagrant@vagrant:~/GeeksForGeeks\$' is shown again at the bottom of the terminal.

Chapter 1 Operating System Overview

4. Concepts in operating systems

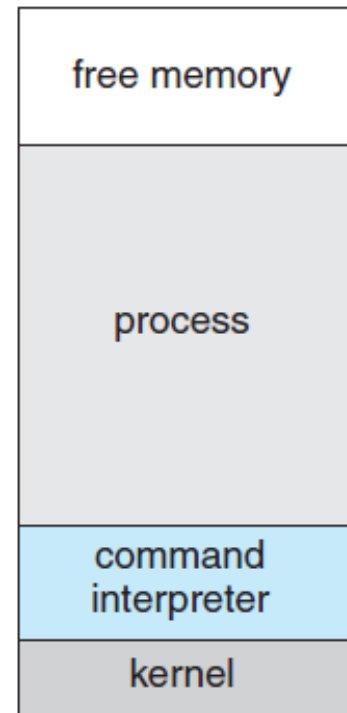
4.3 Shell

Single task environment (MS-DOS)

- Shell waits until the process finishes and then receive new command



(a) At system startup.



(b) Running a program

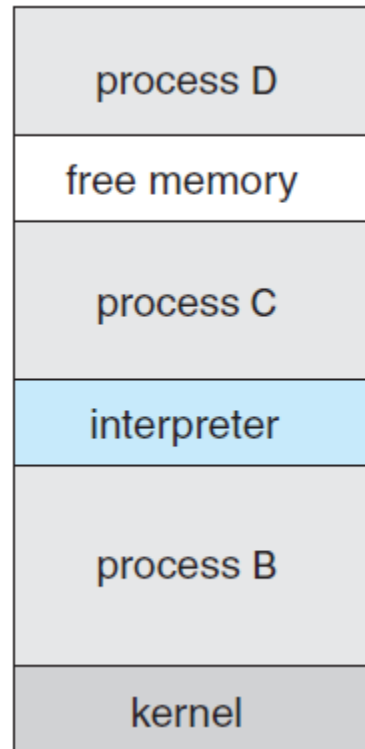
Chapter 1 Operating System Overview

4. Concepts in operating systems

4.3 Shell

multi-tasking system (UNIX, WINXP, . . .)

- After creating and running new process, Shell can receive new command



Chapter 1 Operating System Overview

4. Concepts in operating systems

Concepts in operating systems

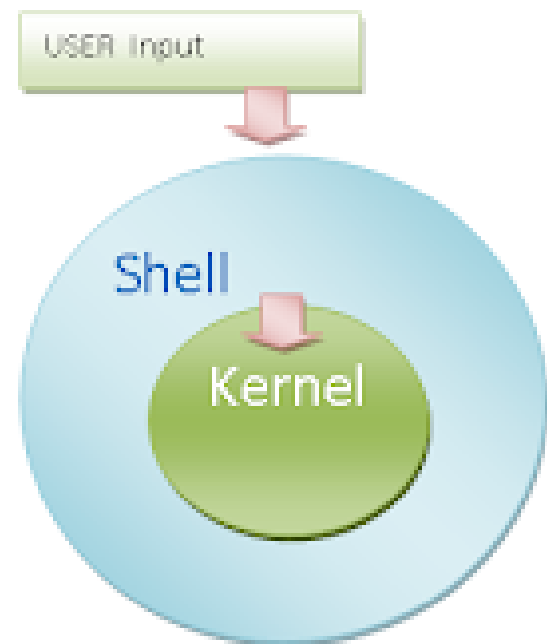
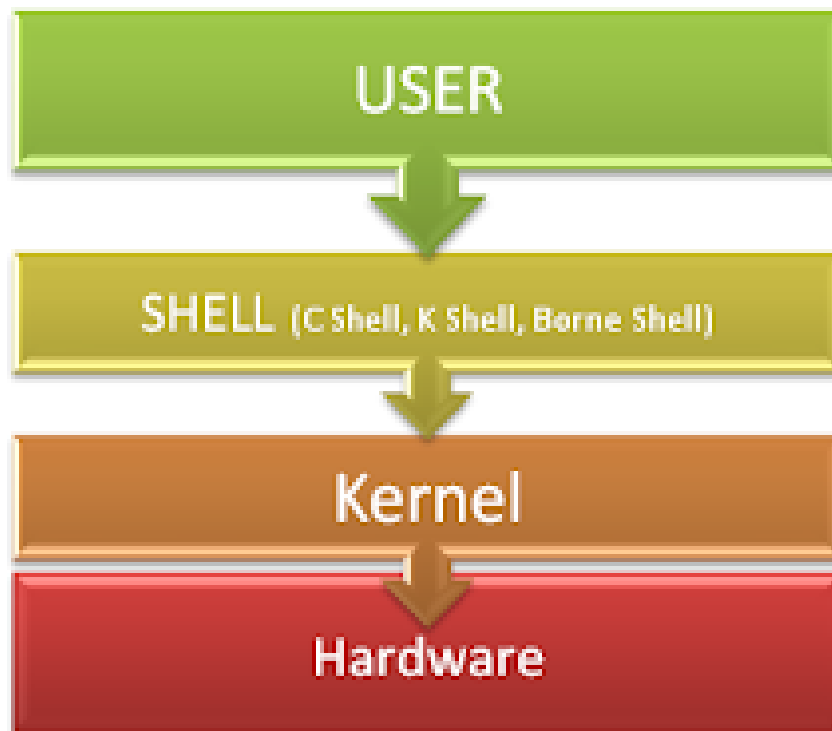
- Process and Thread
- System's resources
- Shell
- Kernel
- System calls

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.4 Kernel

Kernel



Chapter 1 Operating System Overview

4. Concepts in operating systems

4.4 Kernel

Kernel

- A computer program
 - At the lowest level, communicate with the hardware.
 - Control other parts, the core – heart of the OS
 - Loaded first right after the boot loader
 - Translate IO requests from software to a set of CPU instruction
- Hardware protection:
 - User's program directly access -> error notify

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.4 Kernel

Kernel mode

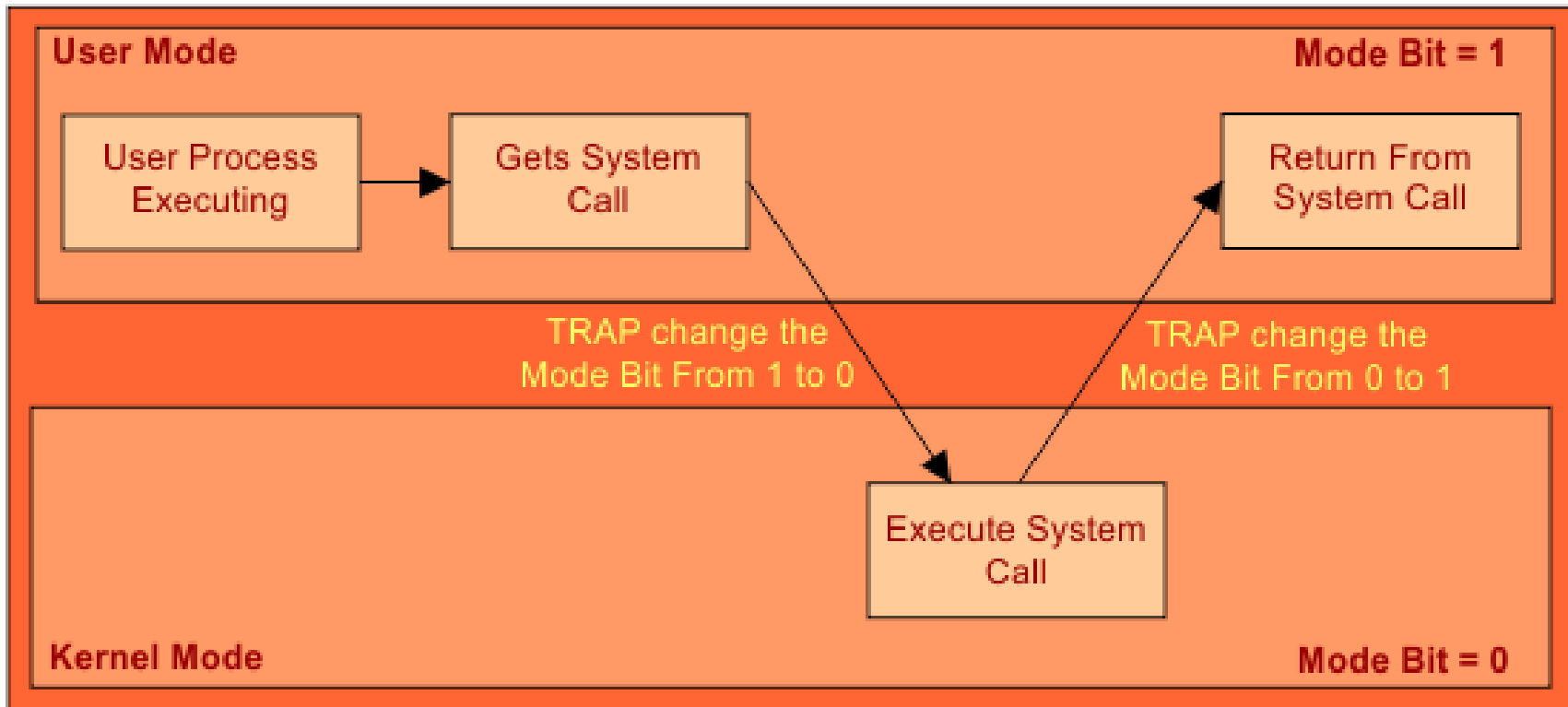
- A privileged mode of operation where the OS has full access to all hardware resources and system components.
 - allows the execution of critical tasks
- Key Characteristics
 - Unlimited Access
 - Execution of Core OS Functions
 - Privileged Instructions
 - Risk of Crashes

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.4 Kernel

Kernel mode and User mode



Chapter 1 Operating System Overview

4. Concepts in operating systems

4.4 Kernel

User mode

- A restricted mode of operation where applications run without direct access to hardware resources.
 - ensure that user programs do not accidentally or maliciously damage the system.
- Key Characteristics
 - Restricted Access
 - System Call Interface
 - Protection

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4. Concepts in operating systems

Concepts in operating systems

- Process
- System's resources
- Shell
- Kernel
- System calls

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.5 System calls

- Provides environment for interacting between user's program and the OS
 - Programs utilize system calls to request services from OS
 - Create, delete, use other software objects operated by the OS
 - Every single system call is corresponding to a library of sub-programs (functions)
- System calls are done in the form of
 - Instructions in low-levels programming languages
 - Interrupt requests (**Int**) in assembly language
 - API functions calls in Windows
- Input parameters for the services and returned results are in special memory areas
 - For example: when making request for an interrupt, the function name is stored in the register AH
 - Int 05 : print to monitor ; Int 13/AH=03h : DISK – WRITE/ DISK SECTOR

Chapter 1 Operating System Overview

4. Concepts in operating systems

4.4 System calls

Example

Func BOOL WINAPI ExitWindowsEx(int uFlags, int dwReason);

uFlags	Shutdown types
EWX_LOGOFF	End process and exit Windows
EWX_POWEROFF	Shutdown system and turn off computer
EWX_REBOOT	Shutdown and restart computer
dwReason	Reason for shutdown

File log_off.c

```
#include <windows.h>
int main(int argc, char *argv[]){
    ExitWindowsEx(EWX_LOGOFF, 0);
    return 0;
}
```

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6. Operating System structures

- System's components
- Operating system's services
- System calls
- Operating system's structures

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components



Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

Process management

- Process: A running program
 - utilize system's resources to complete its task
 - Resources are allocated when process created or while it's running
 - Process terminates, resources are returned
- It is possible for many processes to exist in the system at the same time
 - System process
 - User process

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

Process management

(cont.)

- tasks of OS in process management
 - Create and terminate user's process and system's process
 - Block or re-execute a process
 - Provide mechanism for
 - process synchronization
 - processes' communication
 - controlling deadlock among processes

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

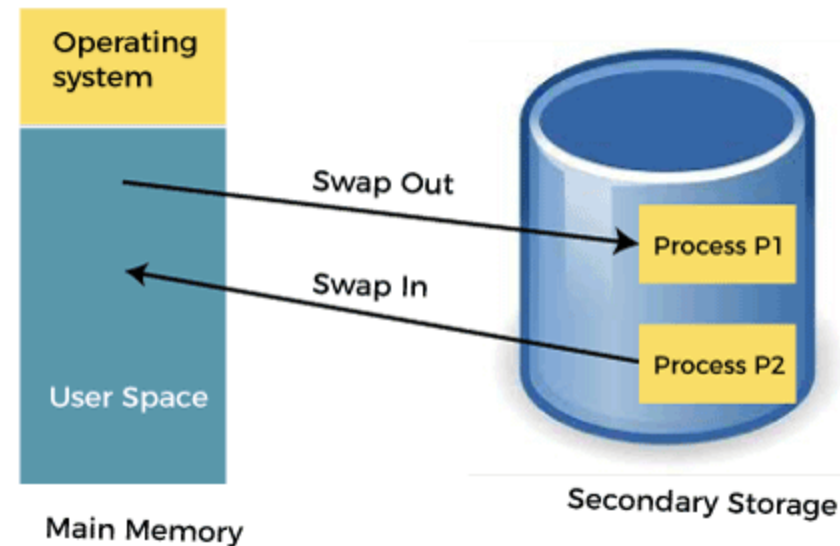
Main memory management

- Main memory: an array of byte(word);
 - Each element has an **address**; where data are accessed by CPU
- To be **executed**, a program must be given an **absolute address** and loaded into main memory.
- When the program is **running**, the **system** accesses instructions and data in main memory.
- To **optimize CPU time** and **computer's speed**, some processes are kept in memory at the same time

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6.1 System's components

Main memory management (cont.)

- Role of OS in main memory management
 - Store information about used areas in memory and who used them
 - Decide which process will be fetched into main memory when the memory is available.
 - Allocate and retrieve memory when it's necessary



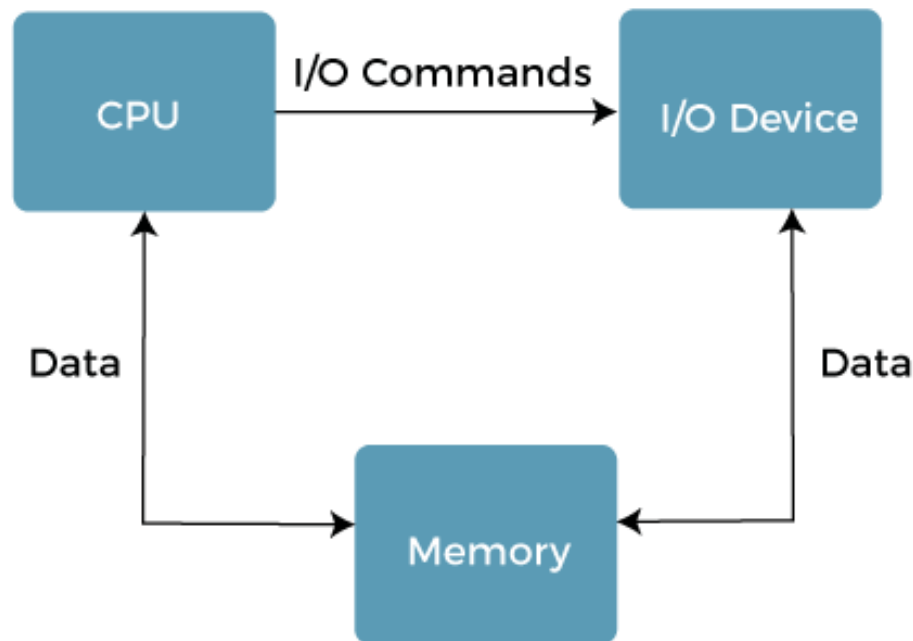
Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

Input-Output system management

- **Objective:** hide physical devices' details from users to help them operate easier.



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6. Operating System structures

6.1 System's components

Input-Output system management

- **Input-Output system management includes**
 - Memory management of buffering, caching, spooling
 - Communicate with device drivers.
 - Controller for special hardware devices. Only device driver understand its associated-device's specific structure

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

File management

- Computer can store information on many types of storage devices
- File: storage unit
- File management task
 - Creates/ deletes a file/directory
 - Provides operations over files and directory
 - Reflects file on secondary storage system
 - Backs up file system on storage devices

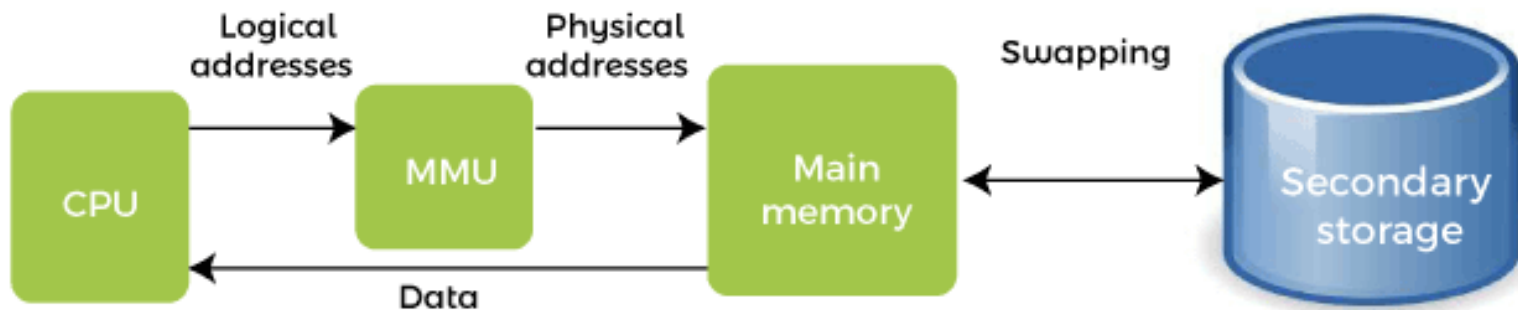
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6. Operating System structures

6.1 System's components

Storage memory management

- Program is stored in secondary memory (magnetic disk) until it's fetched into main memory and executed.
- Disk is utilized for storing data and processed result.



Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

Storage memory management

- Data and result can be stored temporarily on disk: virtual memory
- Role of OS in disk management
 - Unused area management
 - Provide storage area as requested
 - Schedule disk accessing effectively

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

Data transmission system (Distributed system)

- Distributed system combined of set of processor (sym/asym) without common clock and memory. Each processor has a local memory.
- Processor connected via transmission network
- Transmission is performed via protocols (FTP, HTTP...)
- Distributed system allow user to access different resource
- Access to sharing resources will allow
 - Increase computing speed
 - Increase data availability
 - Increase the system reliable

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

System's protection

- **Multi users** operate with the system **at the same time** $\Rightarrow$
Processes must be protected from other processes' activities
- Protection is a controlling mechanism of program or user's access to system or resource
- Protection mechanism will require
 - Distinguish between legal or illegal usage
 - Set imposed controls
 - Provide tools for imposing

Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

User interface

- Carry out user's command.
- Commands are provided for OS 's command controller to
 - Create and manage process
 - Manage main memory and storage memory
 - Access file system
 - Protect
 - Network system
 - ...
- User interface can be command line (DOS, UNIX) or more friendly with graphical interface (Windows, MacOS)

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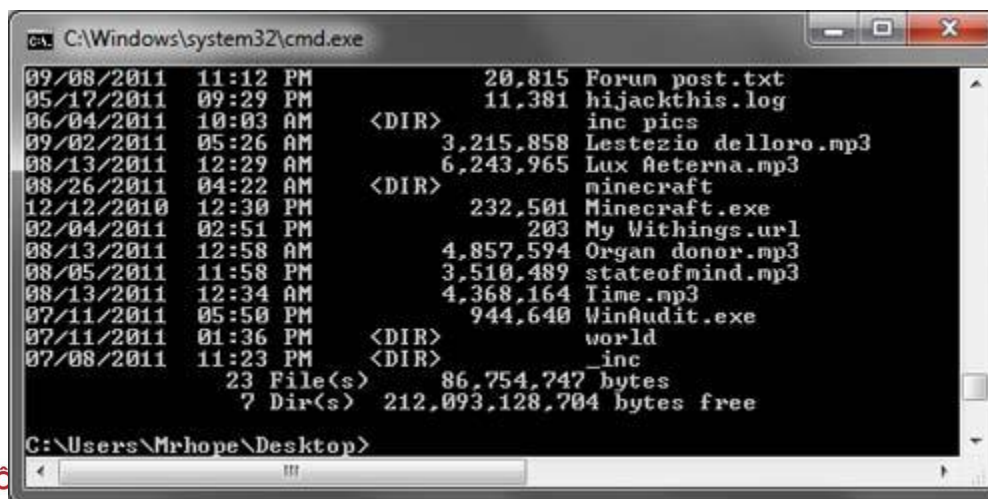
6. Operating System structures

6.1 System's components

User interface

Some forms of human-computer interact

- Command line
 - Simple but organized
 - Do not require complex system specification
 - Easy to add parameter



```
C:\Windows\system32\cmd.exe

09/08/2011  11:12 PM                20,815  Forun post.txt
05/17/2011  09:29 PM                11,381  hijackthis.log
06/04/2011  10:03 AM                <DIR>
09/02/2011  05:26 AM            3,215,858  Lestezio delloro.mp3
08/13/2011  12:29 AM            6,243,965  Lux Aeterna.mp3
08/26/2011  04:22 AM                <DIR>
12/12/2010  12:30 PM            232,501  Minecraft.exe
02/04/2011  02:51 PM                 203  My Withings.url
08/13/2011  12:58 AM            4,857,594  Organ donor.mp3
08/05/2011  11:58 PM            3,510,489  stateofmind.mp3
08/13/2011  12:34 AM            4,368,164  Time.mp3
07/11/2011  05:50 PM            944,640  WinAudit.exe
07/11/2011  01:36 PM                <DIR>
07/08/2011  11:23 PM                <DIR>
                23 File(s)      86,754,747 bytes
                7 Dir(s)    212,093,128,704 bytes free

C:\Users\Mrhope\Desktop>
```

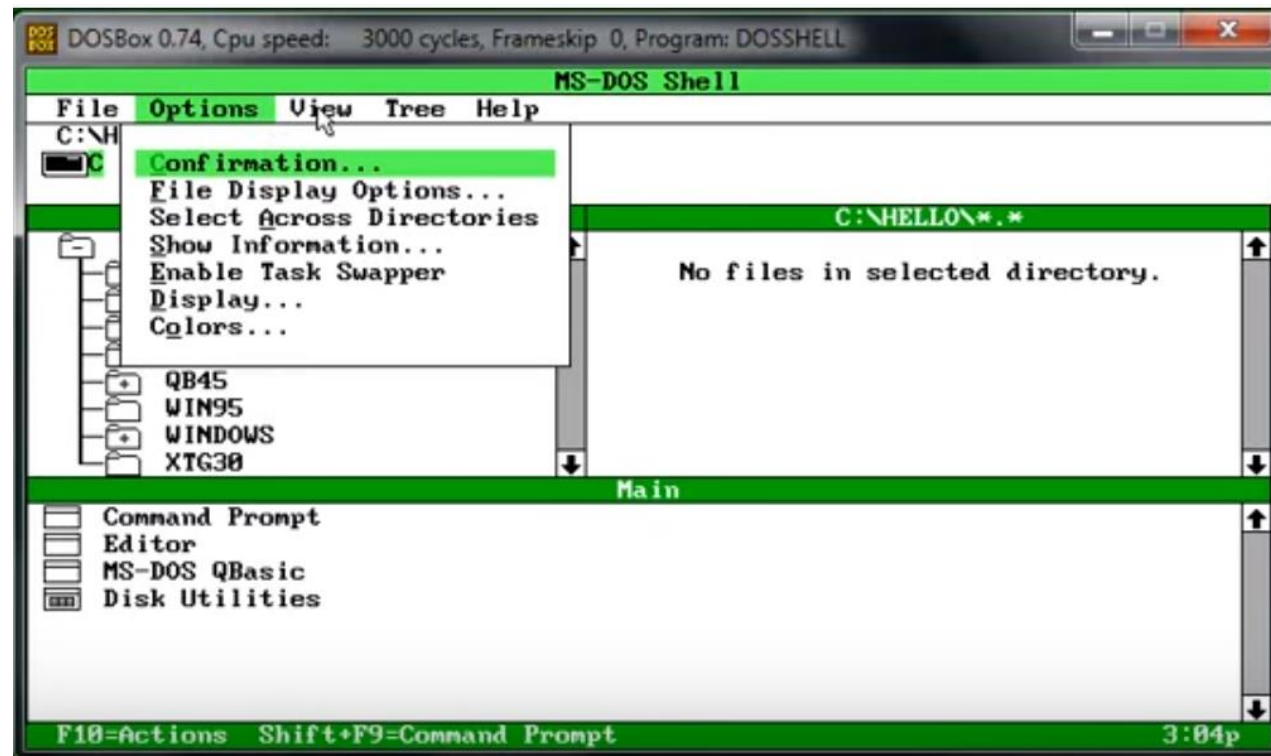
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6. Operating System structures

6.1 System's components

Some forms of human-computer interact (cont 1.)

- Selection table
 - Menu
 - Popup
 - Menu_popup: 2 method: on and onselect



Chapter 1 Operating System Overview

6. Operating System structures

6.1 System's components

Some forms of human-computer interact (cont2.)

- Symbol
window, icon, desktop



Chapter 1 Operating System Overview

6. Operating System structures

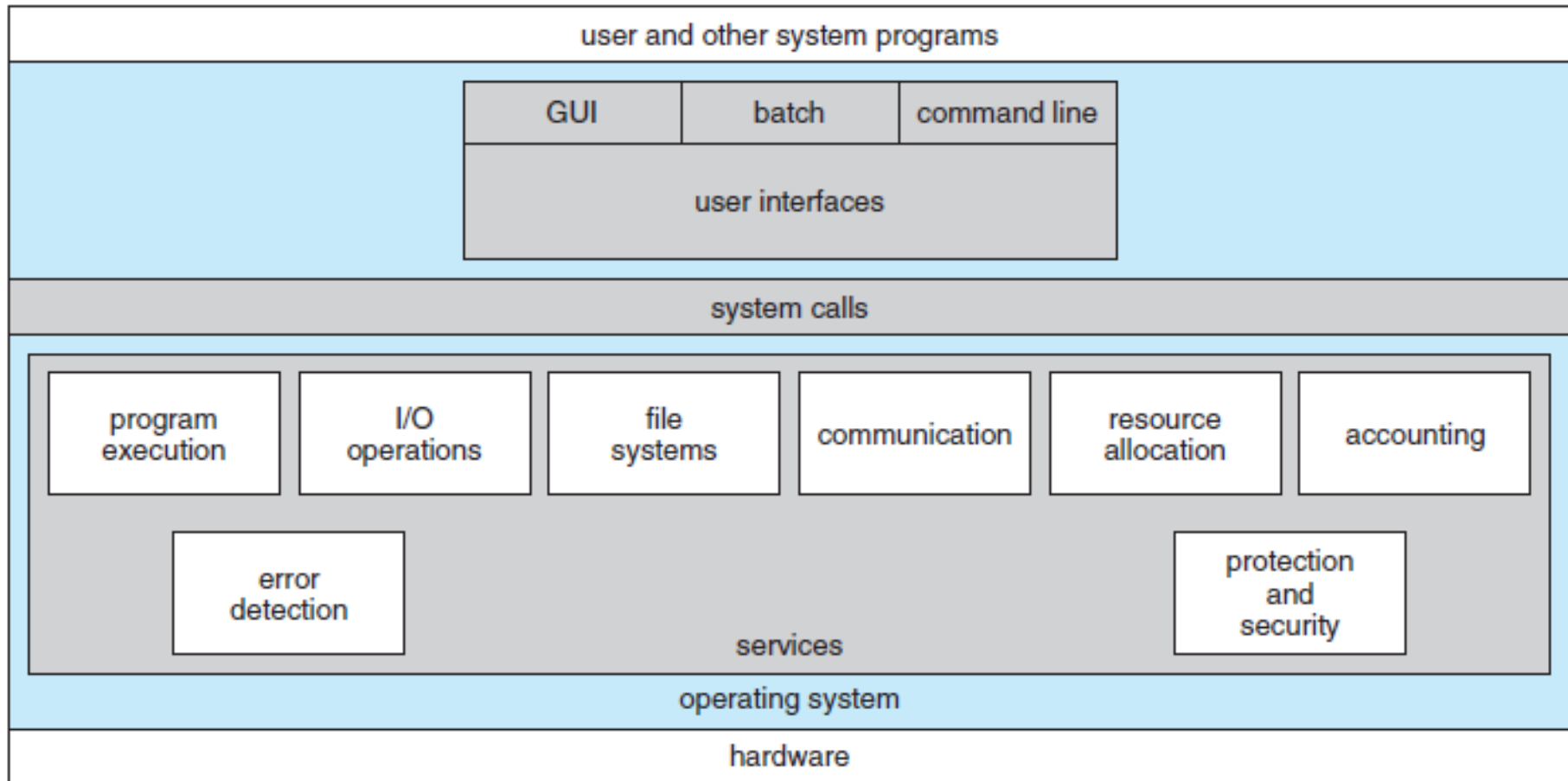
- System's components
- Operating system's services
- System calls
- Operating system's structures

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6. Operating System structures

6.2 Operating system's services

A view of operating system services

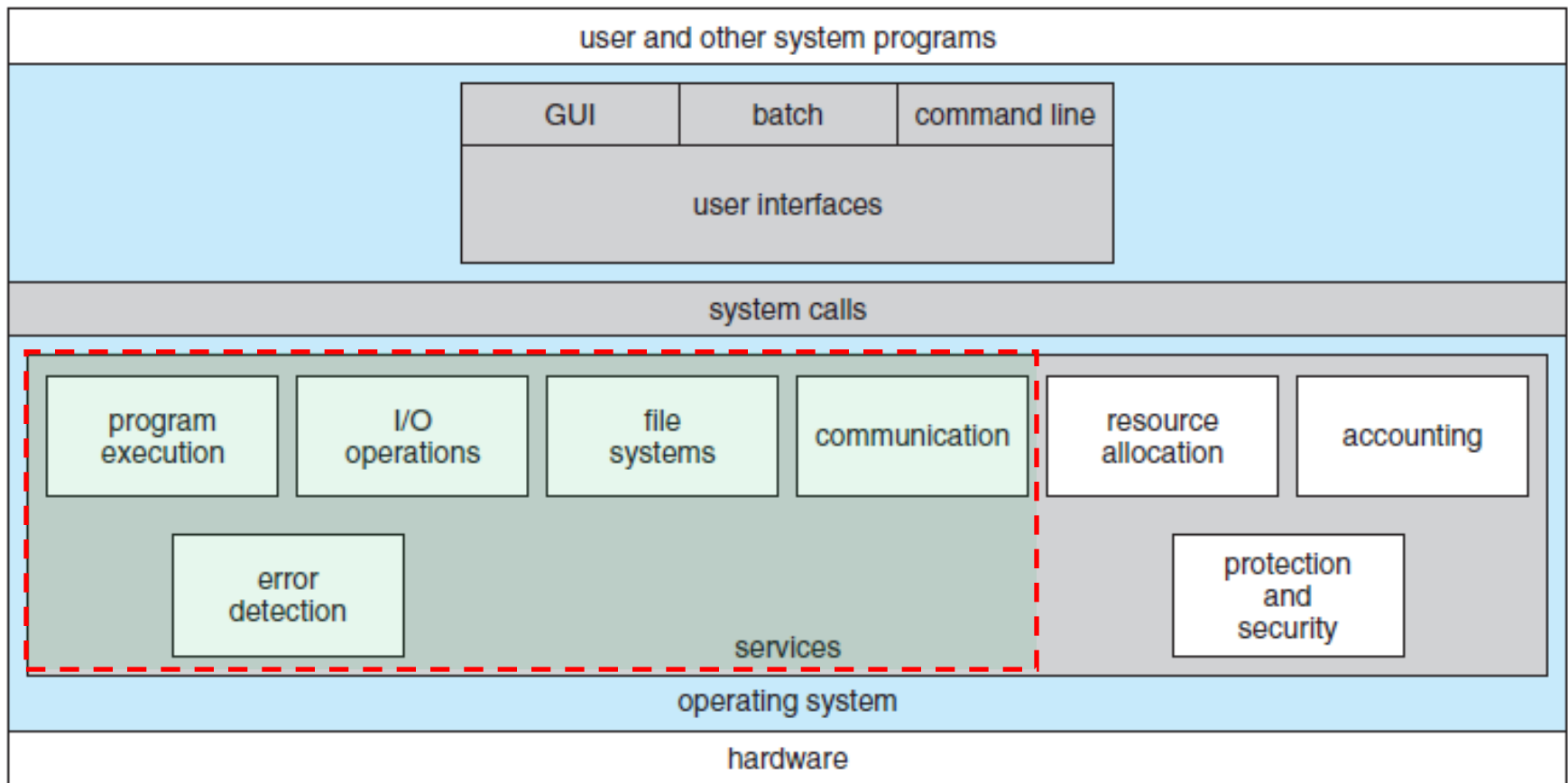


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6. Operating System structures

6.2 Operating system's services

Main and basic services

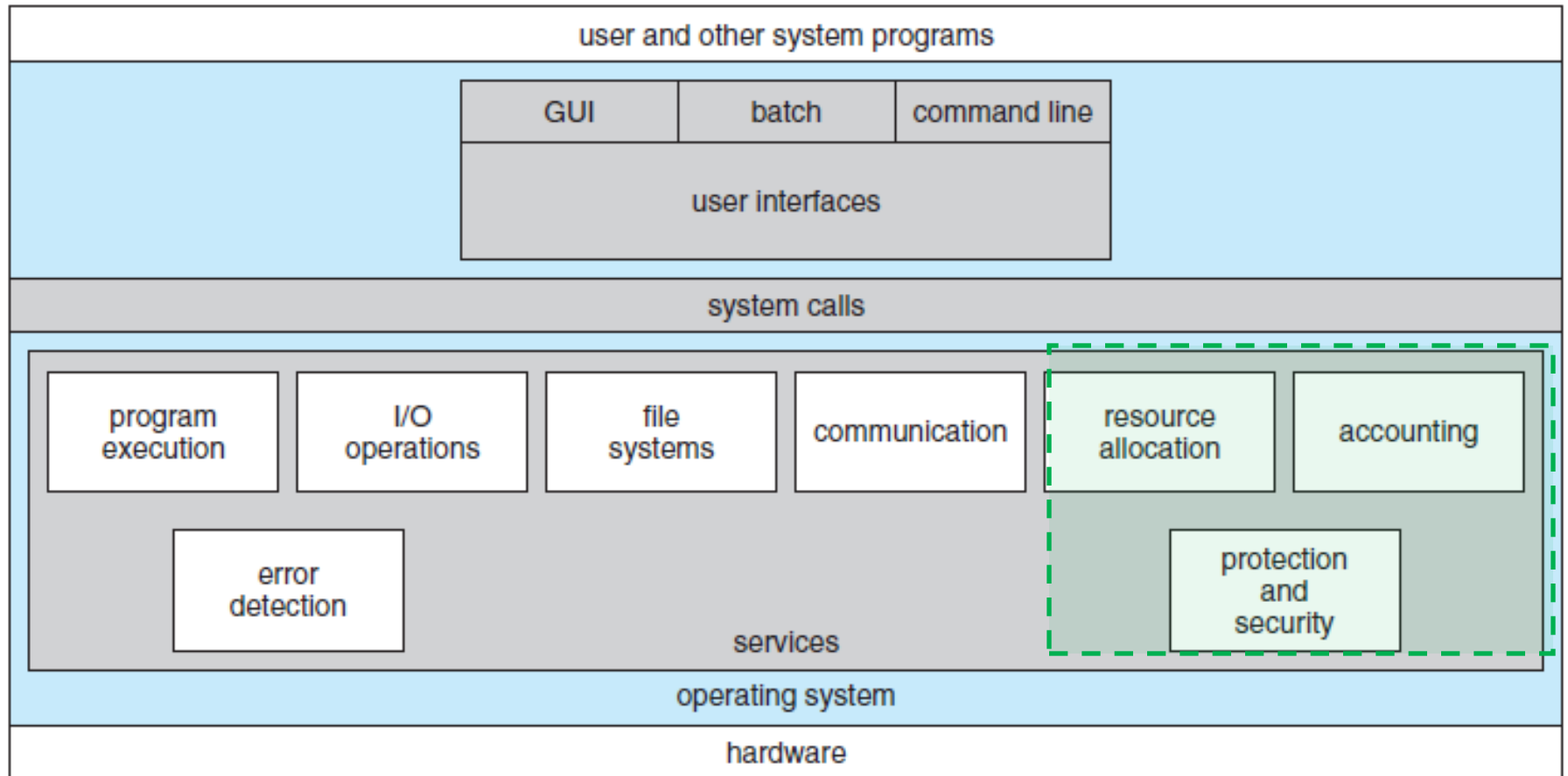


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6. Operating System structures

6.2 Operating system's services

Support services



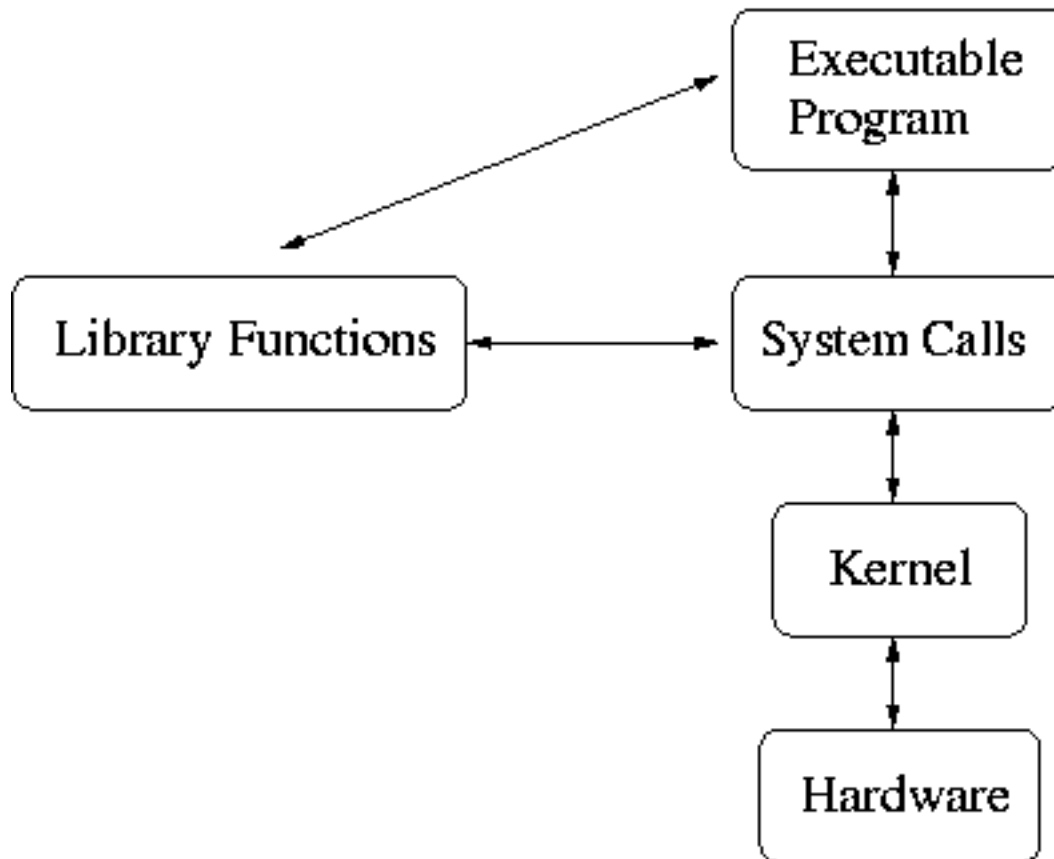
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- System's components
- Operating system's services
- **System calls**
- Operating system's structures

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6.3 System call

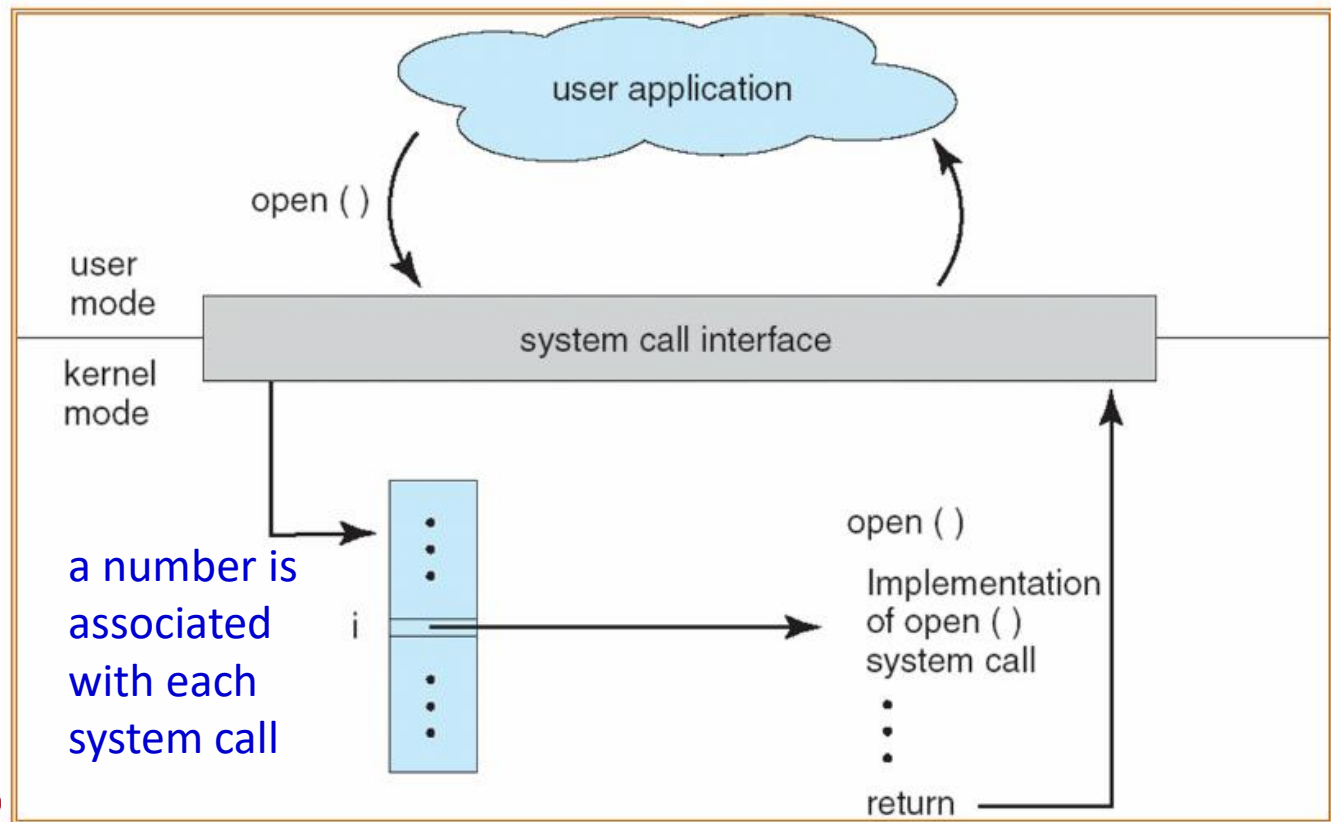
System call



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6. Operating System structures
6.3 System call

System call

Provide an interface between process and operating system



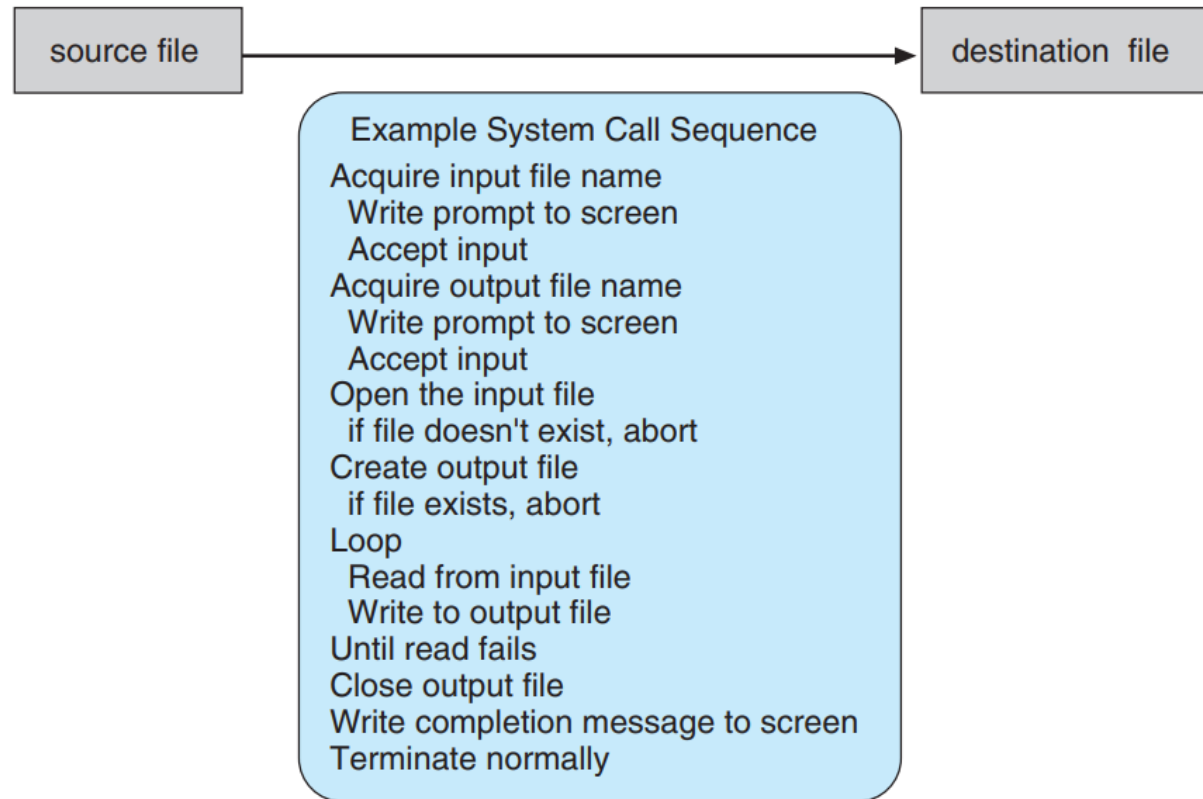
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6. Operating System structures

6.3 System call

System call

Example of how system calls are used in file copying



Chapter 1 Operating System Overview

5. Operating System structures

5.3 System call

System call

- Process management: initialize, terminate process..
- Memory management: allocate and free memory...
- File management: create, delete, read and write file...
- Input Output device management: perform input/output...
- Exchange information with the system. Example get/set time/date...
- Inter process communication

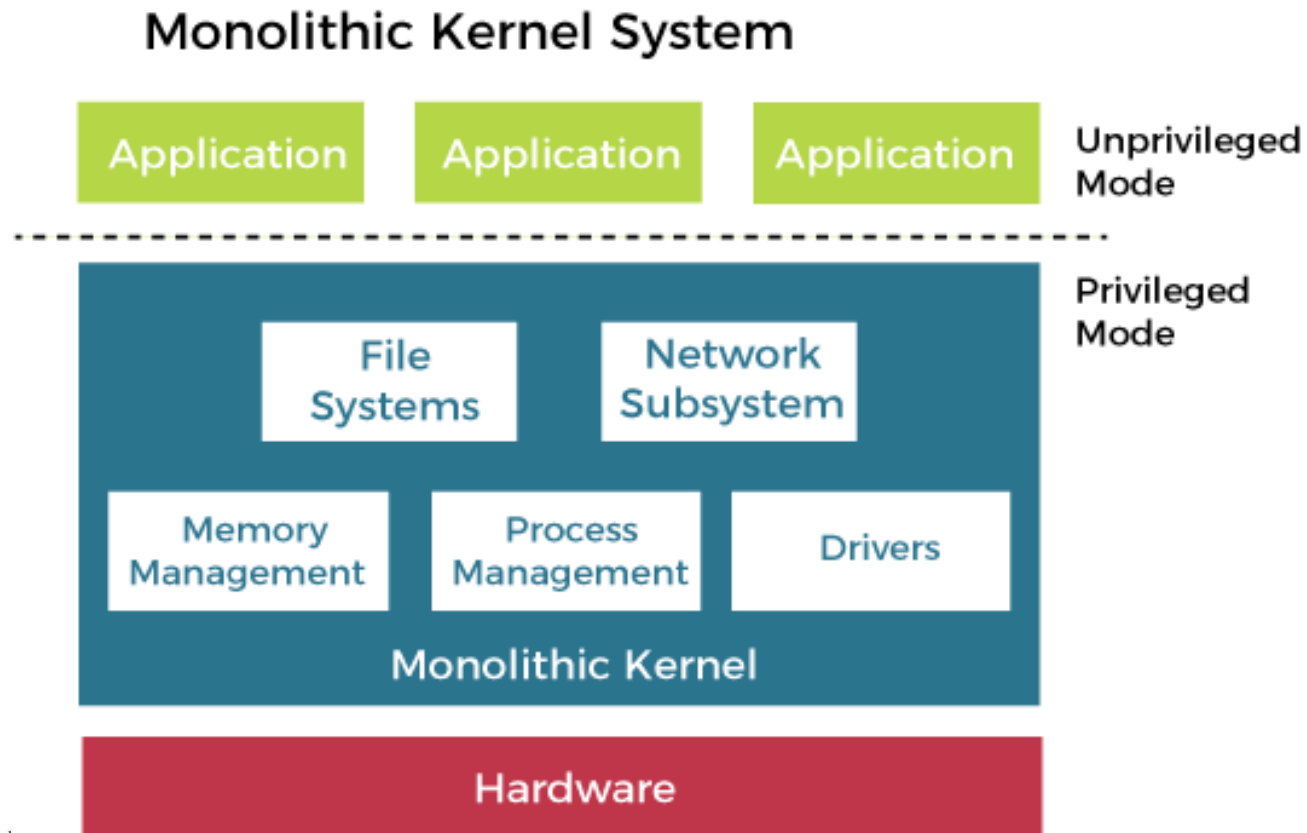
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5. Operating System structures

- System's components
- Operating system's services
- System calls
- Operating system's structures

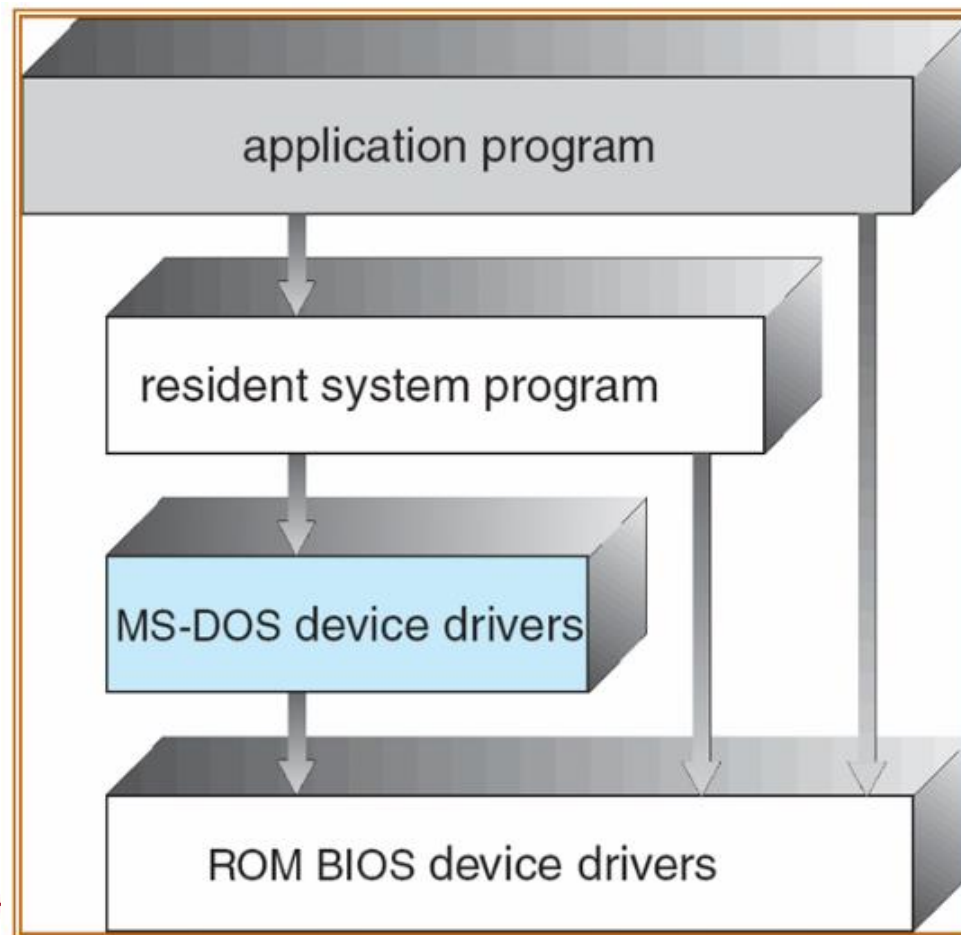
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Monolithic System Architecture



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5.4 Operating system's structures

MS-DOS structure (Silberschatz 2002)

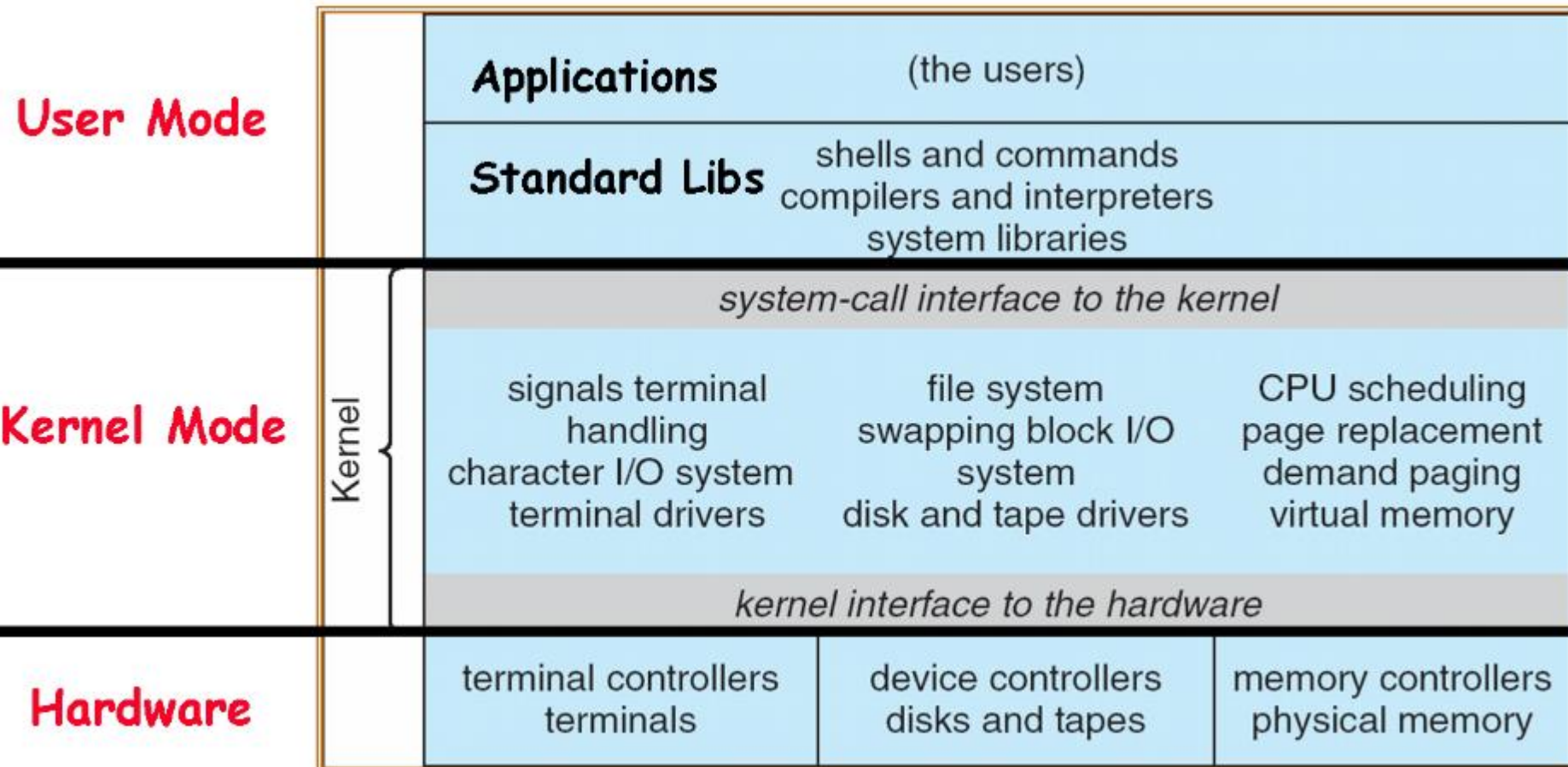


Chapter 1 Operating System Overview

5. Operating System structures

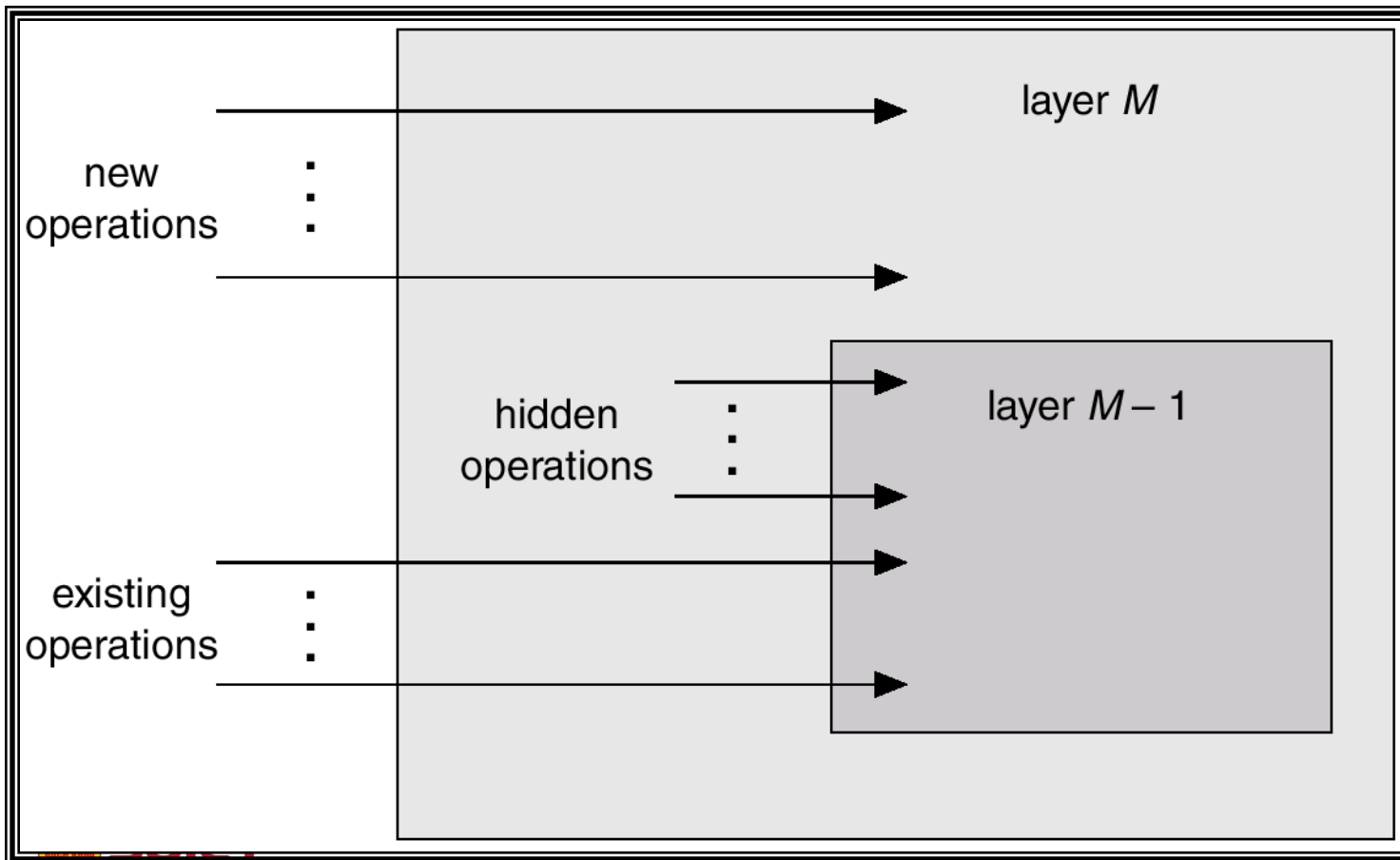
5.4 Operating system's structures

UNIX structure (Silberschatz 2002)



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5.4 Operating system's structures

Layered Approach



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5.4 Operating system's structures

Layered Approach

Ex: Structure of THE (Technische Hogeschool Eindhoven), 1968

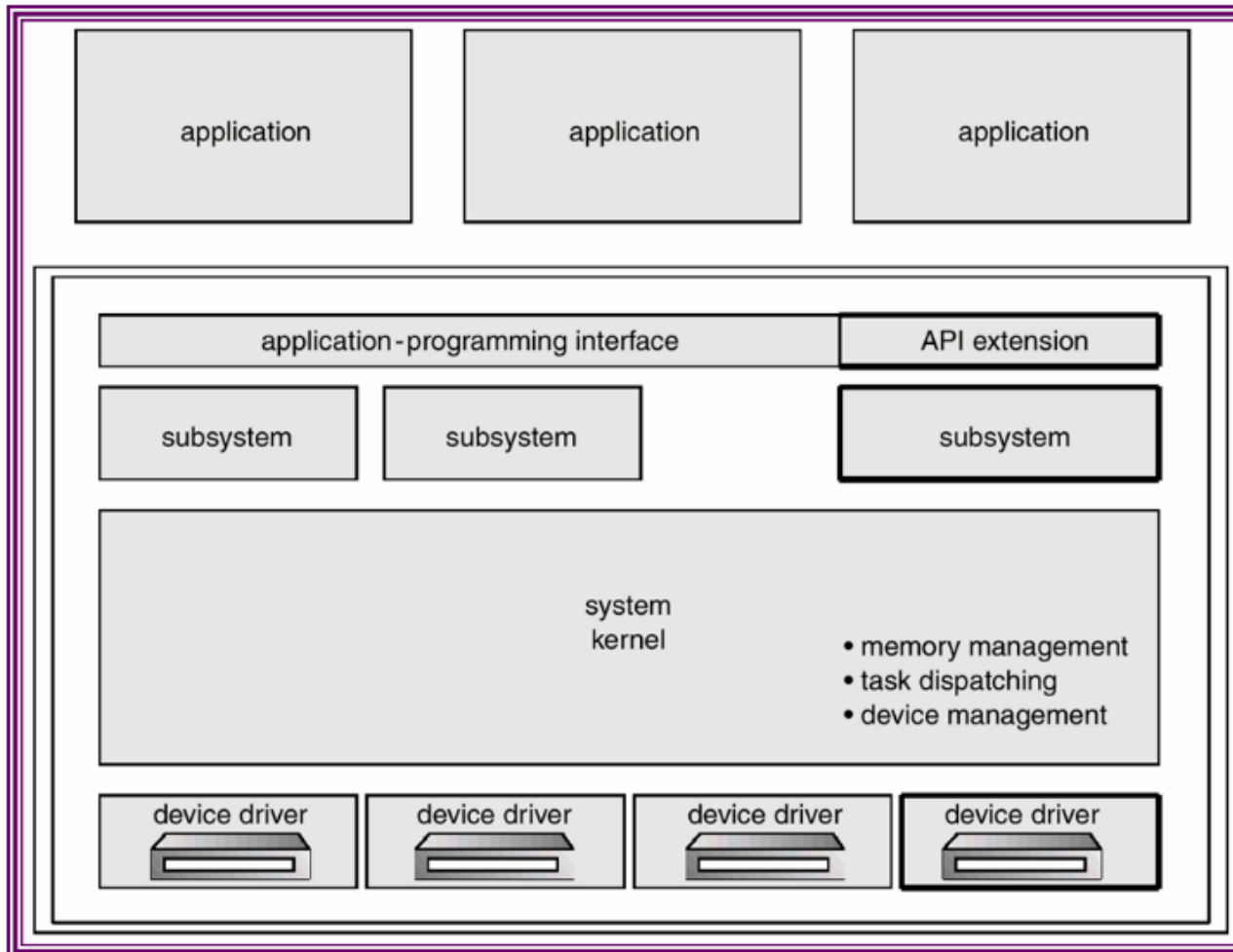
Layer	Function
5	The operator
4	User programs
3	Input/output management
2	Operator-process communication
1	Memory and drum management
0	Processor allocation and multiprogramming

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5. Operating System structures

5.4 Operating system's structures

OS/2 structure (Silberschatz 2002)

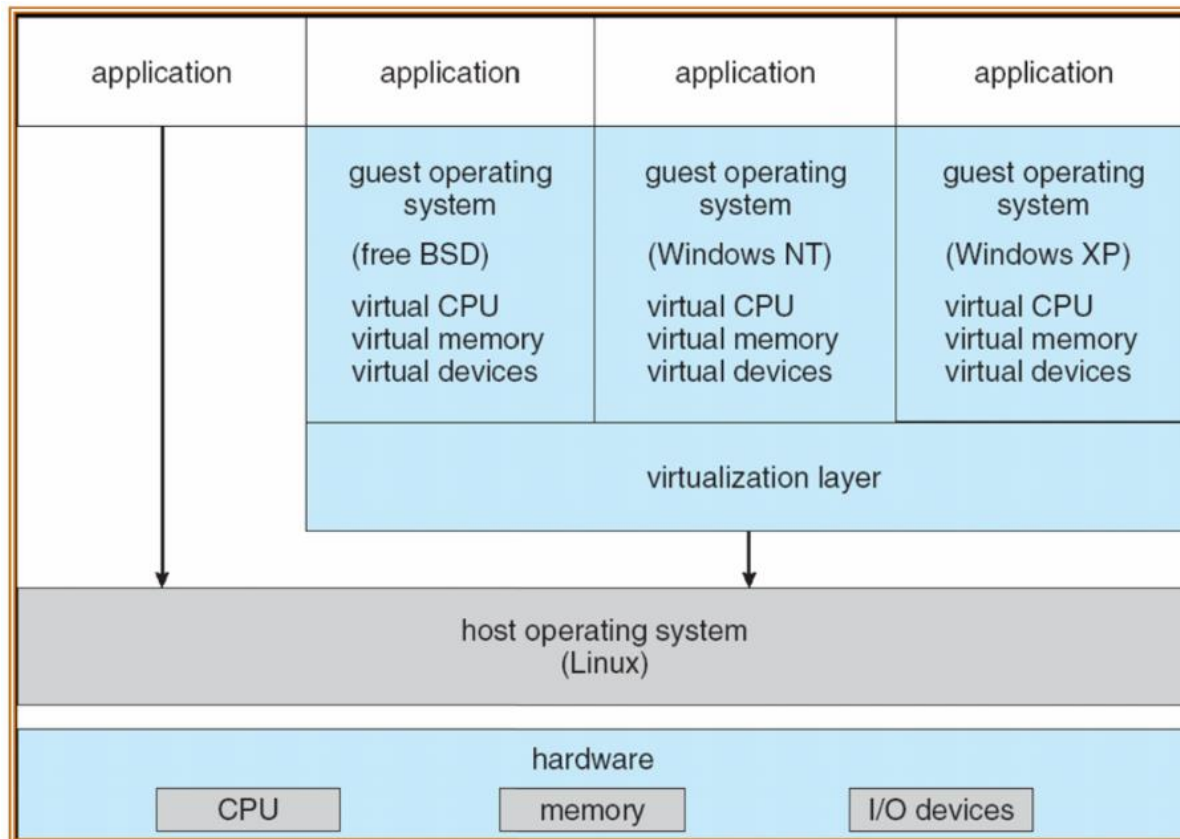


Chapter 1 Operating System Overview

5. Operating System structures

5.4 Operating system's structures

Virtual machine(Silberschatz 2002)



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5. Operating System structures

5.4 Operating system's structures

Microkernel System Structure

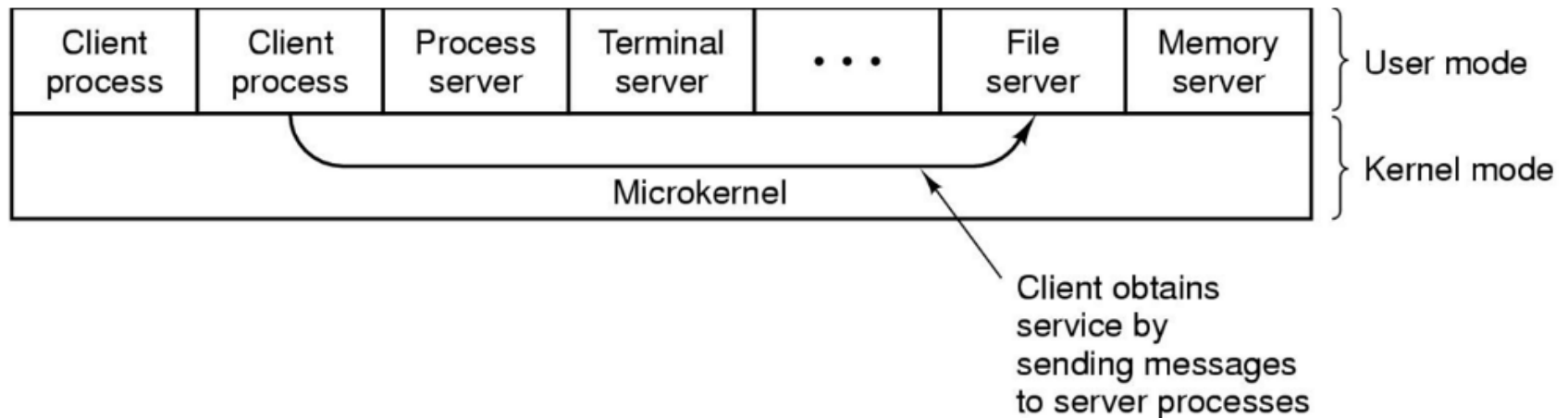
- Moves as much from the kernel into “user” space.
- Communication takes place between user modules using message passing.
- Benefits:
 - easier to extend a microkernel
 - easier to port the operating system to new architectures
 - more reliable (less code is running in kernel mode)
 - more secure

Chapter 1 Operating System Overview

5. Operating System structures

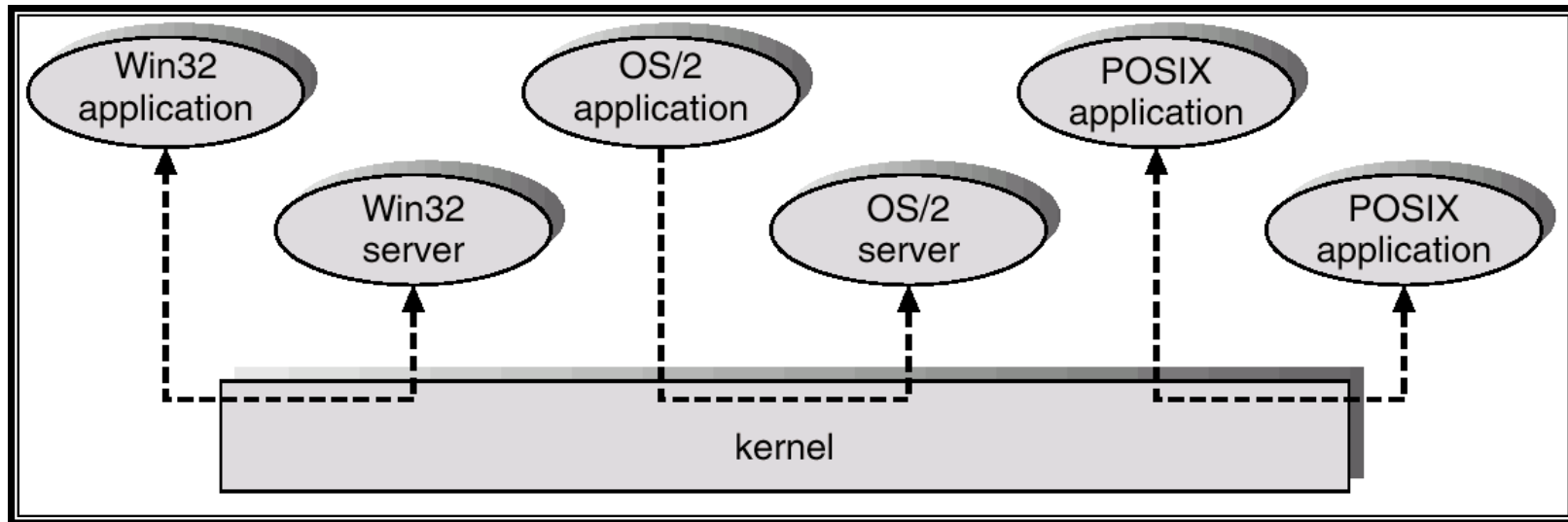
5.4 Operating system's structures

Client-Server Model (Tanenbaum 2001)



Chapter 1 Operating System Overview
5. Operating System structures
5.4 Operating system's structures

Windows NT Client-Server Structure)

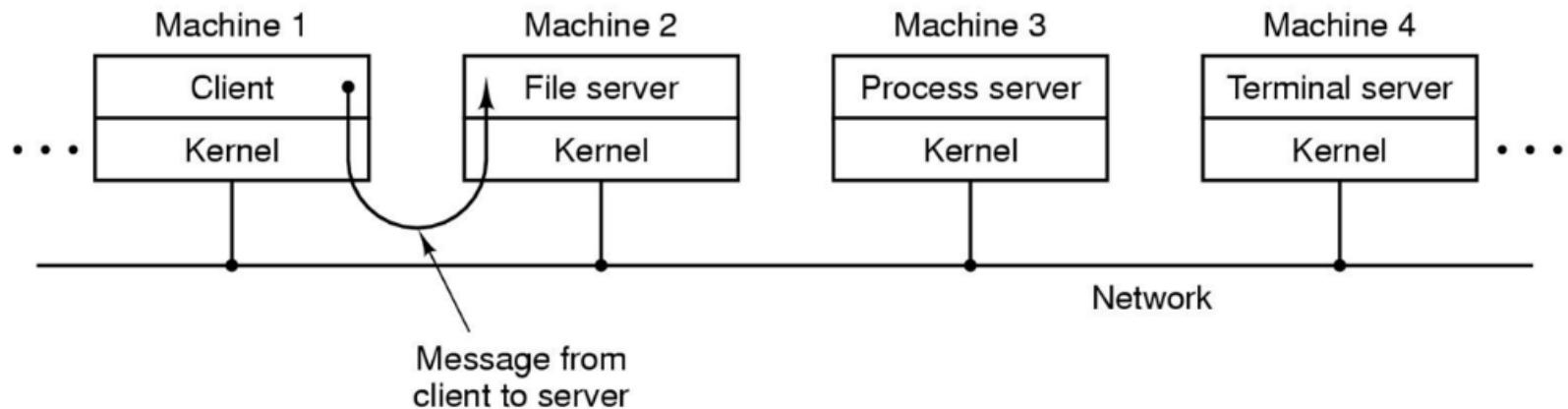


Chapter 1 Operating System Overview

5. Operating System structures

5.4 Operating system's structures

Client-Server model in distributed OS (Tanenbaum 2001)



Chapter 1 Operating System Overview

- ① Operating system Definition
- ② History of Operating Systems
- ③ Definition and Classifications
- ④ Concepts in operating systems
- ⑤ Operating System structures
- ⑥ Basic Properties of Operating Systems
- ⑦ Principles of Operating Systems

Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Basic Properties of Operating Systems

- High reliability
- Secure
- Effectiveness
- General overtime/ Inherit and adaption
- Convenience

Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

High reliability

- Every actions, notifications must be **accurate**
 - Only provide information when it's surely correct
 - **When error occurs** notify and stop the process or let the user decide
 - Require support from device
- Example: C:/>COPY C:/F.TXT A:

High reliability

- Example: C:/>COPY C:/F.TXT A:
 - Check the syntax of command copy
 - Check I/O card (motor, drive accessibility)
 - Check for file F.TXT existence in C drive
 - Check A drive
 - Check if file F.TXT already existed in A drive
 - Check if there is enough space in A
 - Check if the disk is write protection
 - Check written information (if required)

Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Basic Properties of Operating Systems

- High reliability
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Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Security

- Data and programs must be protected
 - No unwanted modification happen in every working mode
 - protected from illegal access
- Different resources have different protection requirements
- Many levels protections with various of tools
- Important for multi tasking system

Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Basic Properties of Operating Systems

- High reliability
- Secure
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Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Effectiveness

- Resources are exploited thoroughly;
- Resource that is limited still able to handle complex requirement.
- The system need to maintain the synchronization;
 - Slow devices do not affect the whole system operation

Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Basic Properties of Operating Systems

- High reliability
- Secure
- Effectiveness
- Generalizable overtime/ Inherit and adaption
- Convenience

Generalizable overtime

- System must be Inheritable
 - Operations, notification can not change
 - If changed: notify and with detailed guide (chkdsk/scandisk)
 - Help keeping and increasing users
- System must have ability to adapt to changes that may happen
- Example: Y2K problem; FAT 12/16/32

Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Basic Properties of Operating Systems

- High reliability
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- General overtime/ Inherit and adaption
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Chapter 1 Operating System Overview

6. Basic Properties of Operating Systems

Convenience

- Easy to use
- Various effective levels
- Have many assisting system

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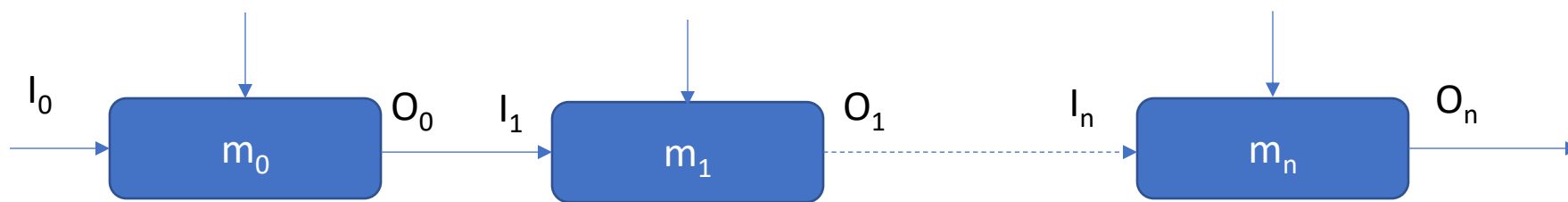
Principles of Operating Systems

- Modular
- Relativity in positioning
- Macroprocessor
- Initialization in the installation
- Functional repetition
- Standard values
- Multi-levels protection

Chapter 1 Operating System Overview

7. Principles of Operating Systems

● Modular principle



The principle of relativity in positioning

- General form of machine instruction:

Instruction_code operand_1 operand_2

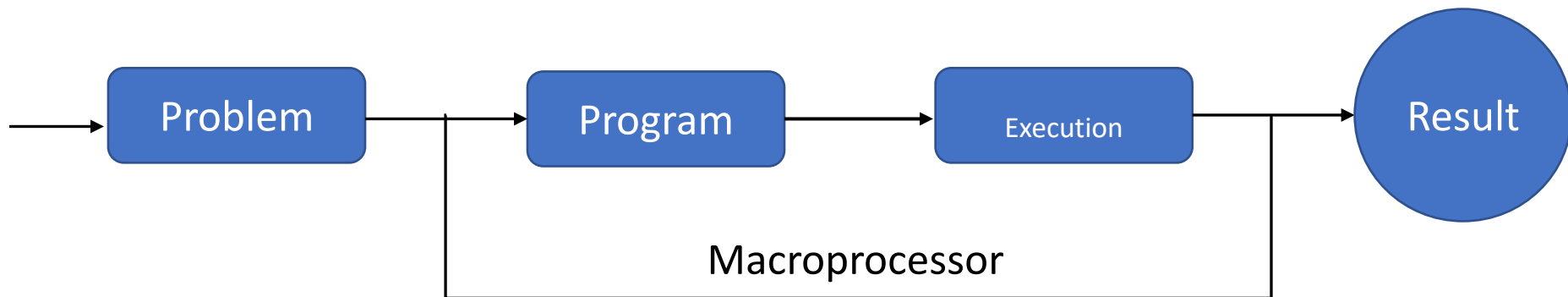
Ex: Mov AX, B

Operand_1, operand_2 can not be absolute address

Chapter 1 Operating System Overview

7. Principles of Operating Systems

The macroprocessor principle



Principle of initialization in the installation

- (Also called generate principle)
- OS is made up of modules in the available module repository to satisfy
 - Machine's configuration
 - User's requirement
- Distinguish **set up** and **install**

Principle of functional repetition

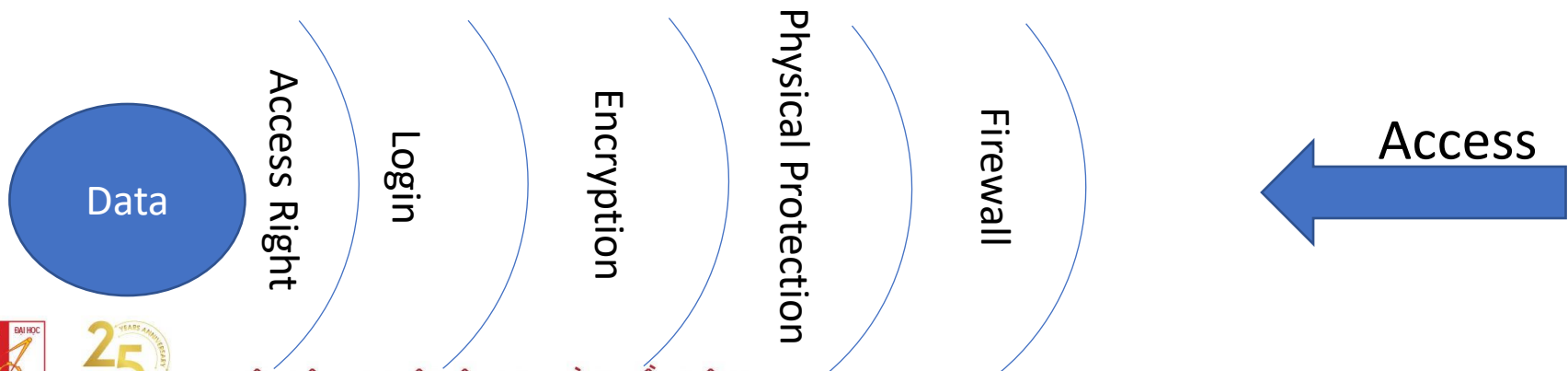
- (1 job can be performed in combinations of different modules
- Make system safer
- Convenient for users' request
- Ex: multiple translation programs for the same language.

Standard values principle

- Increase module flexibility -> module parameterization
- For convenience and safety -> build standard values for modules

Multi-levels protection

- Resources have different characteristics -> different methods of protection
- Multiple layers of protection -> safety
- Help significantly reduce unintentional errors



Chapter 1 Operating System Overview

Summary

- ① **Definition of Operating system**
 - Layering structure of OS
 - OS's functions
- ② **History of Operating system**
 - History of computers
 - History of Operating system
- ③ **Definition and classification of OS**
 - Definitions
 - Classification
- ④ **Basic properties of OS**
 - High reliability
 - Security
 - Effective
 - Generalize overtime
 - Convenience

- ⑤ **Concepts of Operating system**
 - Process and Thread
 - System's resources
 - Shell
 - System calls
- ⑥ **Operating system's structure**
 - OS's components
 - OS's services
 - System calls
 - System's structures
- ⑦ **Principles of Operating System**

